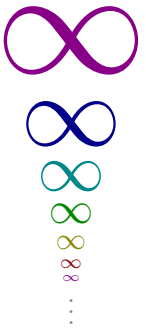
An Infinite Descent into Pure Mathematics

Solution Book



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Chapter 0

Getting started

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Section 0.E

Chapter 0 exercises

0.E.1. The video-sharing website *YouTube* assigns to each video a unique identifier, which is a string of 11 characters from the set

$$\{\texttt{A},\texttt{B},\ldots,\texttt{Z},\texttt{a},\texttt{b},\ldots,\texttt{z},\texttt{0},\texttt{1},\texttt{2},\texttt{3},\texttt{4},\texttt{5},\texttt{6},\texttt{7},\texttt{8},\texttt{9},\texttt{-},\texttt{_}\}$$

This string is actually a natural number expressed in base-64, where the characters in the above set represent the numbers 0 through 63, in the same order—thus `C` represents 2, `c` represents 28, `3` represents 55, and `_` represents 63. According to this schema, find the natural number whose base-64 expansion is `dQw4w9WgXcQ`, and find the base-64 expansion of the natural number 7 159 047 702 620 056 984.

❖ **Solution**

base-64 to base-10:

$$\begin{aligned}\texttt{dQw4w9WgXcQ} =& \texttt{d}\cdot 36^{10}+\texttt{Q}\cdot 36^{9}+\texttt{w}\cdot 36^{8}+\texttt{4}\cdot 36^{7}+\texttt{w}\cdot 36^{6}+\texttt{9}\cdot 36^{5}+\\ & \texttt{W}\cdot 36^{4}+\texttt{g}\cdot 36^{3}+\texttt{X}\cdot 36^{2}+\texttt{c}\cdot 36^{1}+\texttt{Q}\cdot 36^{0}\\ =& 29\cdot 36^{10}+16\cdot 36^{9}+48\cdot 36^{8}+56\cdot 36^{7}+48\cdot 36^{6}+61\cdot 36^{5}+\\ & 22\cdot 36^{4}+32\cdot 36^{3}+23\cdot 36^{2}+28\cdot 36^{1}+16\cdot 36^{0}\\ =& 33\,736\,714\,463\,647\,725\,328\end{aligned}$$

So, the natural number whose base-64 expansion is `dQw4w9WgXcQ` is 33 736 714 463 647 725 328.

base-10 to base-64:

$$\begin{array}{rl|l}
7\,159\,047\,702\,620\,056\,984 & 64\ldots 24 & -Y \\
111\,860\,120\,353\,438\,390 & 64\ldots 54 & -2 \\
1\,747\,814\,380\,522\,474 & 64\ldots 42 & -q \\
27\,309\,599\,695\,663 & 64\ldots 47 & -v \\
426\,712\,495\,244 & 64\ldots 12 & -M \\
6\,667\,382\,738 & 64\ldots 18 & -S \\
104\,177\,855 & 64\ldots 63 & -/ \\
1\,627\,778 & 64\ldots\ 2 & -C \\
25\,434 & 64\ldots 26 & -a \\
397 & 64\ldots 13 & -N \\
6 & 64\ldots\ 6 & -G
\end{array}$$

read remainders backwards

So, the base-64 expansion of the natural number 7 159 047 702 620 056 984 = `GNaC/SMvq2Y`.

0.E.2. Let $a,b,c,d \in \mathbb{Z}$. Under what conditions is $(a+b\sqrt{2})(c+d\sqrt{2})$ an integer?

❖ **Solution**

We know that

$$(a+b\sqrt{2})(c+d\sqrt{2}) = ac+2bd+(ad+bc)\sqrt{2}$$

preview

Since $a,b,c,d \in \mathbb{Z}$, $ac+2bd$ and $ad+bc$ are integers. But $\sqrt{2}$ is irrational, to make $ac+2bd+(ad+bc)\sqrt{2}$ an integer, $(ad+bc)$ must be 0.

Therefore, when $(ad+bc)=0$, $(a+b\sqrt{2})(c+d\sqrt{2})$ is an integer.

0.E.3. Suppose an integer m leaves a remainder of i when divided by 3, and an integer n leaves a remainder of j when divided by 3, where $0 \leqslant i, j < 3$. Prove that $m+n$ and $i+j$ leave the same remainder when divided by 3.

❖ **Proof**

According to the problem statement, we have

$$m = 3k+i \quad \text{and} \quad n = 3l+j$$

where $k,l \in \mathbb{Z}$ and $0 \leqslant i, j < 3$.

$$m+n = (3k+i)+(3l+j) = 3(k+l)+(i+j)$$

Since $3(k+l)$ is multiple of 3, so $3(k+l)+(i+j) \equiv (i+j) \pmod 3$.

Therefore, $m+n$ and $i+j$ leave the same remainder when divided by 3. □

0.E.4. What are the possible remainders of n^2 when divided by 3, where $n \in \mathbb{Z}$?

❖ **Proof**

Let $n \in \mathbb{Z}$. By the division theorem, one of the following must be true for some $k \in \mathbb{Z}$:

$$n = 3k \quad \text{or} \quad n = 3k+1 \quad \text{or} \quad n = 3k+2$$

- Suppose $n = 3k$. Then
$$n^2 = (3k)^2 = 9k^2 = 3 \cdot (3k^2)$$
So n^2 leaves a remainder of 0 when divided by 3.
- Suppose $n = 3k+1$. Then
$$n^2 = (3k+1)^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$$
So n^2 leaves a remainder of 1 when divided by 3.
- Suppose $n = 3k+2$. Then
$$n^2 = (3k+2)^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1$$
So n^2 leaves a remainder of 1 when divided by 3.

In all cases, n^2 leaves a remainder of 0 or 1 when divided by 3. □

✦ **Definition 0.E.5**

A set X is **closed** under an operation $\odot$ if, whenever a and b are elements of X, $a \odot b$ is also an element of X.

In Questions 0.E.6 to 0.E.11, determine which of the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are closed under the operation $\odot$ defined in the question.

0.E.6. $a \odot b = a + b$

0.E.7. $a \odot b = a - b$

0.E.8. $a \odot b = (a-b)(a+b)$

0.E.9. $a \odot b = (a-1)(b-1) + 2(a+b)$

0.E.10. $a \odot b = \dfrac{a}{b^2+1}$

0.E.11. $a \odot b = \dfrac{a}{\sqrt{b^2+1}}$

❖ **Solution**

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
$a \odot b = a + b$	Yes	Yes	Yes	Yes
$a \odot b = a - b$	No	Yes	Yes	Yes
$a \odot b = (a-b)(a+b) = a^2 - b^2$	No	Yes	Yes	Yes
$a \odot b = (a-1)(b-1) + 2(a+b) = (a+1)(b+1)$	Yes	Yes	Yes	Yes
$a \odot b = \dfrac{a}{b^2+1}$	No	No	Yes	Yes
$a \odot b = \dfrac{a}{\sqrt{b^2+1}}$	No	No	No	Yes

Questions 0.E.13 to 0.E.16 concern *algebraic numbers*, defined next.

✦ **Definition 0.E.12** · *Algebraic number*

A complex number α is **algebraic** if $p(\alpha) = 0$ for some nonzero polynomial $p(x)$ over $\mathbb{Q}$.

❖ **Solution**

Questions 0.E.13 to 0.E.16 are existence prooves, which only requires constructing a nonzero polynomial that meets the requirements.

0.E.13. Let x be a rational number. Prove that x is an algebraic number.

❖ **Proof**

Since x is a rational number, so it can be written as $x = \frac{u}{v}$ for some integers u and v with $v \neq 0$. Let nonzero polynomial $p(x)$ over $\mathbb{Q}$ be

$$p(x) = vx - u$$

Then $p(\frac{u}{v}) = v \cdot \frac{u}{v} - u = 0$.

We have shown that there exists a non-zero polynomial $p(x)$ over $\mathbb{Q}$ such that $p(\alpha) = 0$ for any $\alpha \in \mathbb{Q}$. Therefore, $x \in \mathbb{Q}$ is an algebraic number. □

0.E.14. Prove that $\sqrt{2}$ is an algebraic number.

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❖ **Proof**
Let nonzero polynomial $p(x)$ over $\mathbb{Q}$ be

$$p(x) = x^2 - 2$$

Then $p(\sqrt{2}) = (\sqrt{2})^2 - 2 = 0$.

We have shown that there exists a non-zero polynomial $p(x)$ over $\mathbb{Q}$ such that $p(\sqrt{2}) = 0$. Therefore, $\sqrt{2}$ is an algebraic number. □

0.E.15. Prove that $\sqrt{2} + \sqrt{3}$ is an algebraic number.

❖ **Proof**
Let $x = \sqrt{2} + \sqrt{3}$, then

$$\begin{aligned} x^2 = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6} \\ x^2 - 5 = 2\sqrt{6} \\ (x^2 - 5)^2 = (2\sqrt{6})^2 = 24 \\ x^4 - 10x^2 - 1 = 0 \end{aligned}$$

So $\sqrt{2} + \sqrt{3}$ is one root of non-zero polynomial $p(x) = x^4 - 10x^2 - 1$.

We have shown that there exists a non-zero polynomial $p(x)$ over $\mathbb{Q}$ such that $p(\sqrt{2} + \sqrt{3}) = 0$. Therefore, $\sqrt{2} + \sqrt{3}$ is an algebraic number. □

0.E.16. Prove that $x + yi$ is an algebraic number, where x and y are any two rational numbers.

❖ **Proof**
Let $z = x + yi$, then

$$\begin{aligned} z^2 = (x + yi)^2 = x^2 - y^2 + 2xyi \\ z^2 - x^2 + y^2 = 2xyi \\ (z^2 - x^2 + y^2)^2 = (2xyi)^2 = -4x^2y^2 \\ z^4 - 2(x^2 - y^2)z^2 + (x^2 + y^2) = 0 \end{aligned}$$

So $z = x + yi$ is one root of non-zero polynomial $p(z) = z^4 - 2(x^2 - y^2)z^2 + (x^2 + y^2) = 0$.

We have shown that there exists a non-zero polynomial $p(x)$ over $\mathbb{Q}$ such that $p(z) = 0$ for any $x + yi$ where $x, y \in \mathbb{Q}$. Therefore, $z = x + yi$ is an algebraic number. □

Questions 0.E.17 to 0.E.20 concern the *modulus* and *argument* of a complex number.

0.E.17. Let $z \in \mathbb{C}$.

(a) Show that there exist $r, \theta \in \mathbb{R}$ such that $z = r\cos(\theta) + r\sin(\theta)i$.

(b) Show that if $r_1, r_2, \theta_1, \theta_2 \in \mathbb{R}$ are such that $z = r_1\cos(\theta_1) + r_1\sin(\theta_1)i = r_2\cos(\theta_2) + r_2\sin(\theta_2)i$, then $|r_1| = |r_2|$, and if $z \neq 0$, then $\theta_2 = \theta_1 + n \cdot 2\pi$ for some $n \in \mathbb{Z}$.

preview

(c) Conclude that if $z \neq 0$ then there is exactly one $r \in [0,\infty)$ and exactly one $\theta \in (-\pi,\pi]$ such that $z = r\cos(\theta) + r\sin(\theta)i$.

❖ **Proof**

(a) Let $z = x + yi$, where $x, y \in \mathbb{R}$. There are two cases:

- If $z = 0$. Let $r = 0$ and $\theta = 0$, then

$$z = 0\cos(0) + 0\sin(0)i = 0$$

- If $z \neq 0$. Let $r = \sqrt{x^2+y^2} > 0$, then

$$\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1$$

Hence the point $\left(\frac{x}{r}, \frac{y}{r}\right)$ lies on the unit circle, so there exists $\theta \in \mathbb{R}$ such that

$$\cos\theta = \frac{x}{r} \ , \ \sin\theta = \frac{y}{r}$$

Therefore, $z = x + yi = r\cos(\theta) + r\sin(\theta)i$.

In conclusion, there always exist $r, \theta \in \mathbb{R}$ such that $z = r\cos(\theta) + r\sin(\theta)i$.

(b) Assume $r_1, r_2 > 0$, since $z = r_1\cos(\theta_1) + r_1\sin(\theta_1)i = r_2\cos(\theta_2) + r_2\sin(\theta_2)i$, then

$$|z|^2 = r_1^2\cos^2(\theta_1) + r_1^2\sin^2(\theta_1) = r_1^2$$

$$|z|^2 = r_2^2\cos^2(\theta_2) + r_2^2\sin^2(\theta_2) = r_2^2$$

So $r_1^2 = r_2^2$, this gives $|r_1| = |r_2|$.

Since $|r_1| = |r_2|$, we have

$$\cos(\theta_1) + \sin(\theta_1)i = \cos(\theta_2) + \sin(\theta_2)i$$

then

$$\cos(\theta_1) = \cos(\theta_2) \ \text{ and } \ \sin(\theta_1) = \sin(\theta_2)$$

therefore $\theta_1 = \theta_2 + n \cdot 2\pi$.

(c) Let $z \neq 0$

- **Existence:** Let $r = |z| > 0$. By part (a), there exists $\theta_0 \in \mathbb{R}$ such that

$$z = r(\cos(\theta_0) + \sin(\theta_0)i)$$

By part (b), there exists some $n \in \mathbb{Z}$ such that

$$\theta = \theta_0 + n \cdot 2\pi \in (-\pi, \pi]$$

Then

$$z = r\cos(\theta) + r\sin(\theta)i$$

where $r \in [0, +\infty)$ and $\theta \in (-\pi, \pi]$.

preview

- **Uniqueness:** Suppose

$$z = r_1\cos(\theta_1) + r_1\sin(\theta_1)i = r_2\cos(\theta_2) + r_2\sin(\theta_2)i$$

where $r_1, r_2 \in [0, +\infty)$ and $\theta_1, \theta_2 \in (-\pi, \pi]$.
By part (b), we have

$$|z| = r_1 = r_2 > 0 \text{ and } \theta_1 = \theta_2 + n \cdot 2\pi$$

Since $\theta_1, \theta_2 \in (-\pi, \pi]$, we have

$$\theta_1 - \theta_2 \in (-2\pi, 2\pi)$$

so the only multiple of 2π in this interval is 0. Hence $\theta_1 = \theta_2$. □

Definition 0.E.18 · *Modulus and argument of a complex number*
Let $z \in \mathbb{C}$. The **modulus** $|z|$ and (**principal**) **argument** $\arg(z)$ of z are the elements $|z| \in [0, \infty)$ and $\arg(z) \in (-\pi, \pi]$ such that

$$z = |z|\cos(\arg(z)) + |z|\sin(\arg(z))i$$

where we set $\arg(z) = 0$ if $z = 0$.

Note that the modulus and argument exist and are unique by Question 0.E.17.

0.E.19. Let $z_1, z_2 \in \mathbb{C}$, and for $k = 1, 2$ define $r_k = |z_k|$ and $\theta_k = \arg(z_k)$.

Prove that $z_1 z_2 = r_1 r_2 \cos(\theta_1 + \theta_2) + r_1 r_2 \sin(\theta_1 + \theta_2)i$, and therefore express $|z_1 z_2|$ and $\arg(z_1 z_2)$ in terms of $r_1, r_2, \theta_1, \theta_2$.

Proof
Let $z_1 = r_1\cos(\theta_1) + r_1\sin(\theta_1)i$ and $z_2 = r_2\cos(\theta_2) + r_2\sin(\theta_2)i$. Then

$$\begin{aligned} z_1 z_2 &= \big(r_1\cos(\theta_1) + r_1\sin(\theta_1)i\big)\big(r_2\cos(\theta_2) + r_2\sin(\theta_2)i\big) \\ &= r_1 r_2\cos(\theta_1)\cos(\theta_2) - r_1 r_2\sin(\theta_1)\sin(\theta_2) + r_1 r_2\cos(\theta_1)\sin(\theta_2)i + r_1 r_2 sin(\theta_1)\cos(\theta_2)i \\ &= r_1 r_2\big(\cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)\big) + r_1 r_2\big(\sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2)\big)i \\ &= r_1 r_2\cos(\theta_1 + \theta_2) + r_1 r_2\sin(\theta_1 + \theta_2)i \end{aligned}$$

So, $|z_1 z_2| = r_1 r_2$ and $\arg(z_1 z_2) = \theta_1 + \theta_2$. □

0.E.20. Repeat Question 0.E.19, but for $\dfrac{z_1}{z_2}$ instead of $z_1 z_2$.

preview

❖ **Proof**

Let $z_1 = r_1\cos(\theta_1) + r_1\sin(\theta_1)i$ and $z_2 = r_2\cos(\theta_2) + r_2\sin(\theta_2)i$. Then

$$\begin{aligned}
\frac{z_1}{z_2} &= \frac{r_1\cos(\theta_1) + r_1\sin(\theta_1)i}{r_2\cos(\theta_2) + r_2\sin(\theta_2)i} \\
&= \frac{\big(r_1\cos(\theta_1) + r_1\sin(\theta_1)i\big)\big(r_2\cos(\theta_2) - r_2\sin(\theta_2)i\big)}{\big(r_2\cos(\theta_2) + r_2\sin(\theta_2)i\big)\big(r_2\cos(\theta_2) - r_2\sin(\theta_2)i\big)} \\
&= \frac{r_1r_2\cos(\theta_1)\cos(\theta_2) + r_1r_2\sin(\theta_1)\sin(\theta_2) + r_1r_2\sin(\theta_1)\cos(\theta_2)i - r_1r_2\cos(\theta_1)\sin(\theta_2)i}{\big(r_2\cos(\theta_2)\big)^2 - \big(r_2\sin(\theta_2)i\big)^2} \\
&= \frac{r_1r_2\big(\cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)\big) + r_1r_2\big(\sin(\theta_1)\cos(\theta_2) - \cos(\theta_1)\sin(\theta_2)\big)i}{r_2^2\cos^2(\theta_2) + r_2^2\sin^2(\theta_2)} \\
&= \frac{r_1r_2\cos(\theta_1-\theta_2) + r_1r_2\sin(\theta_1-\theta_2)i}{r_2^2} \\
&= \frac{r_1}{r_2}\cos(\theta_1-\theta_2) + \frac{r_1}{r_2}\sin(\theta_1-\theta_2)i
\end{aligned}$$

So, $\left|\frac{z_1}{z_2}\right| = \frac{r_1}{r_2}$ and $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2$. □

0.E.21. Prove a general version of the quadratic formula for quadratic polynomials over $\mathbb{C}$.

❖ **Proof**

Let $p(x) = ax^2 + bx + c$, where $a, b, c \in \mathbb{C}$ and $a \neq 0$. We prove that the roots of p are given by

$$\frac{-b \pm \sqrt{\Delta}}{2a}$$

where $\sqrt{\Delta} \in \mathbb{C}$ satisfies $\Delta = b^2 - 4ac$.

$$\begin{aligned}
p(x) &= ax^2 + bx + c \\
&= a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) \\
&= a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2}\right]
\end{aligned}$$

Let $\Delta = b^2 - 4ac$, and choose $\sqrt{\Delta} \in \mathbb{C}$, then

$$\begin{aligned}
p(x) &= a\left[\left(x + \frac{b}{2a}\right)^2 - \left(\frac{\sqrt{\Delta}}{2a}\right)^2\right] \\
&= a\left(x + \frac{b}{2a} + \frac{\sqrt{\Delta}}{2a}\right)\left(x + \frac{b}{2a} - \frac{\sqrt{\Delta}}{2a}\right)
\end{aligned}$$

Therefore, $p(x) = 0$ if and only if

$$x + \frac{b}{2a} + \frac{\sqrt{\Delta}}{2a} = 0 \quad \text{or} \quad x + \frac{b}{2a} - \frac{\sqrt{\Delta}}{2a} = 0$$

preview

Thus, the roots of $p(x)$ are

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}, \quad \sqrt{\Delta} = \sqrt{b^2 - 4ac} \in \mathbb{C}$$

True–False questions

In Questions 0.E.22 to 0.E.29, determine (with proof) whether the statement is true or false.

0.E.22. Every integer is a natural number.

❖ **Solution**
False

Because $\mathbb{N} \subset \mathbb{Z}$, not all integers are natural numbers. For example, -1 is an integer but is not a natural number.

0.E.23. Every integer is a rational number.

❖ **Solution**
True

Because $\mathbb{Z} \subset \mathbb{Q}$, any integer p can be written as $\frac{p}{1}$, so all integers are rational numbers.

0.E.24. Every integer divides zero.

❖ **Solution**
True

Because $\forall n \in \mathbb{Z}, 0 = n \cdot 0$, according to the Definition 0.33, every integer divides zero.

0.E.25. Every integer divides its square.

❖ **Solution**
True

Because $\forall n \in \mathbb{Z}, n^2 = n \cdot n$, According to the Definition 0.33, every integer divides its square.

0.E.26. The square of every rational number is a rational number.

❖ **Solution**
True

Every rational number can be represented as a ratio of two integers $\frac{p}{q}, p, q \in \mathbb{Z}$. Its square is $\frac{p^2}{q^2}$, which is a rational number.

0.E.27. The square root of every positive rational number is a rational number.

prev

❖ **Solution**
False

We can give a counterexample: 2 is a positive rational number, but its square root $\sqrt{2}$ is an irrational number.

0.E.28. When an integer a is divided by a positive integer b, the remainder is always less than a.

❖ **Solution**
False

According to the division theorem, $a = qb + r$, where $q, r \in \mathbb{Z}$ and the remainder $0 \leqslant r < |b|$. If $a < 0$, then r is greater than a. If $0 \leqslant a < b$, then $r = a$.

0.E.29. Every quadratic polynomial over $\mathbb{R}$ with no real roots has two distinct non-real roots.

❖ **Solution**
True

By Question 0.E.21, if $\Delta > 0$, there are two distinct real roots; If $\Delta = 0$, there is one real root; If $\Delta < 0$, there are no real roots, but there are two distinct non-real roots.

Always–Sometimes–Never questions

In Questions 0.E.30 to 0.E.38, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

0.E.30. Let $n, b_1, b_2 \in \mathbb{N}$ with $1 < n < b_1 < b_2$. Then the base-b_1 expansion of n is equal to the base-b_2 expansion of n.

❖ **Solution**
Always

Given $1 < n < b_1 < b_2$, we have $n < b_1$ and $n < b_2$. In any base $b > n$, the integer n is represented by a single digit equal to n itself (since $0 \leqslant n < b$). Therefore, the base-b_1 expansion of n is the digit n, and the base-b_2 expansion of n is also the digit n. Hence they are identical.

0.E.31. Let $n, b_1, b_2 \in \mathbb{N}$ with $1 < b_1 < b_2 < n$. Then the base-b_1 expansion of n is equal to the base-b_2 expansion of n.

❖ **Solution**
Never

Given $1 < b_1 < b_2 < n$, we have $b_1 < n$ and $b_2 < n$. The integer n can be represented as

$$n = a_n \cdot b_1^n + a_{n-1} \cdot b_1^{n-1} + \cdots + a_1 \cdot b_1 + a_0$$

$$n = a'_n \cdot b_2^n + a'_{n-1} \cdot b_2^{n-1} + \cdots + a'_1 \cdot b_2 + a'_0$$

Because $b_1 \neq b_2$, the expansion of n in base b_1 and b_2 are not the same.

preview

0.E.32. Let $a, b, c \in \mathbb{Z}$ and suppose that a divides c and b divides c. Then ab divides c.

❖ **Solution**
Sometimes

We can give a counterexample: $a = 3, b = 6, c = 12$. Then a divides c and b divides c, but ab does not divide c.

0.E.33. Let $a, b, c \in \mathbb{Z}$ and suppose that a divides c and b divides c. Then ab divides c^2.

❖ **Solution**
Always

According to the problem statement, we have

$$c = ma \quad \text{and} \quad c = nb \quad \text{where} \quad m, n \in \mathbb{Z}$$

Then $c^2 = ma \cdot nb = (mn) \cdot (ab)$, which is a multiple of ab.

Therefore, ab divides c^2.

0.E.34. Let $x, y \in \mathbb{Q}$ and let $a, b, c, d \in \mathbb{Z}$ with $cy + d \neq 0$. Then $\dfrac{ax+b}{cy+d} \in \mathbb{Q}$.

❖ **Solution**
Always

$x, y \in \mathbb{Q}$, x, y can be represented as $x = \dfrac{p}{q}$ and $y = \dfrac{r}{s}$, where $p, q, r, s \in \mathbb{Z}$. Then

$$\frac{ax+b}{cy+d} = \frac{\frac{ap}{q} + b}{\frac{cr}{s} + d} = \frac{\frac{ap+bq}{q}}{\frac{cr+ds}{s}} = \frac{s(ap+bq)}{q(cr+ds)}$$

Because $a, b, c, d, p, q, r, s \in \mathbb{Z}$, $\dfrac{ax+b}{cy+d}$ is a rational number.

0.E.35. Let $\dfrac{a}{b}$ be a rational number. Then $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$.

❖ **Solution**
Sometimes

We can give a counterexample: $\dfrac{\sqrt{2}}{\sqrt{2}} = 1$ is a rational number, but $\sqrt{2}$ is not a integer, even not a rational number.

0.E.36. Let $x \in \mathbb{R}$ and assume that $x^2 \in \mathbb{Q}$. Then $x \in \mathbb{Q}$.

preview

❖ **Solution**
Sometimes

We can give a counterexample: Let $x = \sqrt{2}$, $x^2 = 2$ is a rational number, but $\sqrt{2}$ is not a rational number.

0.E.37. Let $x \in \mathbb{R}$ and assume that $x^2 + 1 \in \mathbb{Q}$ and $x^5 + 1 \in \mathbb{Q}$. Then $x \in \mathbb{Q}$.

❖ **Solution**
Always

Given that $x^2 + 1 \in \mathbb{Q}$ and $x^5 + 1 \in \mathbb{Q}$, we can immediately deduce that $x^2 \in \mathbb{Q}$ and $x^5 \in \mathbb{Q}$.

Let $x^2 = a$ and $x^5 = b$.

- If $a = 0$, then $x = 0$. Since 0 is a rational number, $x \in \mathbb{Q}$ is true.
- If $a \neq 0$, then $x = \dfrac{x^5}{x^4} = \dfrac{x^5}{(x^2)^2} = \dfrac{b}{a^2}$. Since $b \in \mathbb{Q}$, $a^2 \in \mathbb{Q}$, and $a^2 \neq 0$, $x \in \mathbb{Q}$ is true.

Therefore, $x \in \mathbb{Q}$ is always true.

0.E.38. Let $p(x) = ax^2 + bx + c$ be a polynomial with $a, b, c \in \mathbb{R}$ and $a \neq 0$, and suppose that $u + vi$ be a complex root of $p(x)$ with $v \neq 0$. Then $u - vi$ is a root of $p(x)$.

❖ **Solution**
Always

By Question 0.E.21, $p(x)$ has two roots

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}, \quad \sqrt{\Delta} = \sqrt{b^2 - 4ac} \in \mathbb{C}$$

If $u + vi$ is one complex root of $p(x)$, we can deduce that $\Delta < 0$, then $\sqrt{\Delta}$ can be written as $\sqrt{-\Delta}i$. Thus

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-b \pm \sqrt{-\Delta}i}{2a}, \quad \Delta < 0$$

Let $u = \dfrac{-b}{2a}$ and $v = \dfrac{\sqrt{-\Delta}}{2a}$, the roots can be written as

$$x = u \pm vi$$

Therefore, $u - vi$ is a root of $p(x)$.

preview

Part I

Core concepts

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Chapter 1

Logical structure

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Section 1.E

Chapter 1 exercises

1.E.1. For fixed $n \in \mathbb{N}$, let p represent the proposition 'n is even', let q represent the proposition 'n is prime' and let r represent the proposition '$n = 2$'. For each of the following propositional formulae, translate it into plain English and determine whether it is true for all $n \in \mathbb{N}$, true for some values of n and false for some values of n, or false for all $n \in \mathbb{N}$.

(a) $(p \wedge q) \Rightarrow r$

(b) $q \wedge (\neg r) \Rightarrow (\neg p)$

(c) $((\neg p) \vee (\neg q)) \vee (\neg r)$

(d) $(p \wedge q) \wedge (\neg r)$

❖ **Solution**

(a) If n is even and n is prime, then $n = 2$. This is true for all $n \in \mathbb{N}$.

(b) If n is prime and $n \neq 2$, then n is not even. This is true for all $n \in \mathbb{N}$.

(c) Either n is not even, or n is not prime, or $n \neq 2$. This is true for all $n \in \mathbb{N}$.

(d) n is even and n is prime, and $n \neq 2$. This is false for all $n \in \mathbb{N}$.

1.E.2. For each of the following plain English statements, translate it into a symbolic propositional formula. The propositional variables in your formulae should represent the simplest propositions that they can.

(a) Guinea pigs are quiet, but they're loud when they're hungry.

(b) It doesn't matter that 2 is even, it's still a prime number.

(c) $\sqrt{2}$ can't be an integer because it is an irrational number.

❖ **Solution**

(a) Let p represent the proposition 'guinea pigs are quiet'.
Let q represent the proposition 'guinea pigs are loud'.
Let r represent the proposition 'guinea pigs are hungry'.
Then the statement can be represented by the propositional formula $p \wedge (r \Rightarrow q)$.

(b) Let p represent the proposition '2 is even'.
Let q represent the proposition '2 is a prime number'.
Then the statement can be represented by the propositional formula $(p \vee \neg p) \Rightarrow q$.

(c) Let p represent the proposition '$\sqrt{2}$ is an integer'.
Let q represent the proposition '$\sqrt{2}$ is an irrational number'.
Then the statement can be represented by the propositional formula $q \Rightarrow \neg p$.

1.E.3. Let p and q be propositions, and assume that $p \Rightarrow (\neg q)$ is true and that $(\neg q) \Rightarrow p$ is false. Which of the following are true, and which are false?

preview

(a) q being false is necessary for p to be true.

(b) q being false is sufficient for p to be true.

(c) p being true is necessary for q to be false.

(d) p being true is sufficient for p to be false.

❖ **Solution**

(a) True. "q being false is necessary for p to be true" means that $p \Rightarrow (\neg q)$. Since $p \Rightarrow (\neg q)$ is true, it follows that this statement is true.

(b) False. "q being false is sufficient for p to be true" means that $(\neg q) \Rightarrow p$. Since $(\neg q) \Rightarrow p$ is false, it follows that this statement is false.

(c) False. "p being true is necessary for q to be false" means that $(\neg q) \Rightarrow p$. Since $(\neg q) \Rightarrow p$ is false, it follows that this statement is false.

(d) True. "p being true is sufficient for q to be false." means that $p \Rightarrow (\neg q)$. Since $p \Rightarrow (\neg q)$ is true, it follows that this statement is true.

In Questions 1.E.4 to 1.E.7, use the definitions of the logical operators in Section 1.1 to describe what steps should be followed in order to prove the propositional formula in the question; the letters p, q, r and s are propositional variables.

1.E.4. $(p \wedge q) \Rightarrow (\neg r)$

❖ **Solution**

To prove $(p \wedge q) \Rightarrow (\neg r)$, we need to assume that $p \wedge q$ is true and then show that $\neg r$ is true.

Since $p \wedge q$ is true, by the definition of conjunction, we have p is true and q is true. We need to use the fact that p and q are both true to derive r is false or a contradiction if we assume that r is true. If we can derive r is false or a contradiction from the assumption that r is true, then it follows that $\neg r$ must be true.

1.E.5. $(p \vee q) \Rightarrow (r \Rightarrow s)$

❖ **Solution**

To prove $(p \vee q) \Rightarrow (r \Rightarrow s)$, we need to assume that $p \vee q$ is true and then show that $r \Rightarrow s$ is true.

Since $p \vee q$ is true, by the definition of disjunction, we have either p is true or q is true. We need to use the fact that at least one of p or q is true to derive that if r is true, then s must be true.

To show that $r \Rightarrow s$ is true, we need to assume that r is true and then show that s is true.

1.E.6. $(p \Rightarrow q) \Leftrightarrow (\neg p \Rightarrow \neg q)$

❖ **Solution**

We know that original proposition $p \Rightarrow q$ is logically equivalent to its contrapositive $\neg q \Rightarrow \neg p$. Therefore, we can rewrite the proposition in this question as

$$(p \Rightarrow q) \Leftrightarrow (q \Rightarrow p)$$

pre

$$[(p \Rightarrow q) \Rightarrow (q \Rightarrow p)] \wedge [(q \Rightarrow p) \Rightarrow (p \Rightarrow q)]$$

This means that $p \Leftrightarrow q$. To prove $p \Leftrightarrow q$, we need to show that $p \Rightarrow q$ and $q \Rightarrow p$ are both true. To show that $p \Rightarrow q$ is true, we need to assume that p is true and then show that q is true. To show that $q \Rightarrow p$ is true, we need to assume that q is true and then show that p is true.

1.E.7. $(p \wedge (\neg q)) \vee (q \wedge (\neg p))$

❖ **Solution**

$(p \wedge (\neg q)) \vee (q \wedge (\neg p))$ equivalent to $p \oplus q$ (exclusive OR). To prove $p \oplus q$, we need to show that either p is true and q is false, or q is true and p is false. To show that p is true and q is false, we need to assume that p is true and then show that q is false. To show that q is true and p is false, we need to assume that q is true and then show that p is false.

1.E.8. Find a statement in plain English, involving no variables at all, that is equivalent to the logical formula $\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}, (a < b \Rightarrow \exists c \in \mathbb{R}, [a < c < b \ \wedge \ \neg(c \in \mathbb{Q})])$. Then prove this statement, using the structure of the logical formula as a guide.

❖ **Solution**

Between any two distinct rational numbers, there exists an irrational number.

Proof
This is an existential statement. Let $a, b \in \mathbb{Q}$ be arbitrary, we need to construct a real number c such that $a < c < b$ and $c \notin \mathbb{Q}$.

Let $c = a + \dfrac{b-a}{\sqrt{2}}$.

- Show $a < c < b$
  Since $a < b$, we have $b - a > 0$, so $\dfrac{b-a}{\sqrt{2}} > 0$. Adding a to both sides gives $c = a + \dfrac{b-a}{\sqrt{2}} > a$.
  Since $\sqrt{2} > 1$, we have $\dfrac{b-a}{\sqrt{2}} < b - a$. Adding a to both sides gives $c = a + \dfrac{b-a}{\sqrt{2}} < a + (b - a) = b$.
  Therefore, $a < c < b$.
- Show $c \notin \mathbb{Q}$
  Since $\sqrt{2}$ is irrational and a, b are rational, $c = a + \dfrac{b-a}{\sqrt{2}}$ is irrational.

Thus we have exhibited an irrational number c strictly between a and b. Since a, b were arbitrary rational numbers with $a < b$, the statement holds for all such pairs. □

1.E.9. Find a purely symbolic logical formula that is equivalent to the following statement, and then prove it: "*No matter which integer you may choose, there will be an integer greater than it.*"

❖ **Solution**

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, (y > x)$$

preview

Proof
Let $x \in \mathbb{Z}$ be arbitrary. We need to find an integer y such that $y > x$. Let $y = x+1$. Since x is an integer, y is also an integer. Moreover, $y = x+1 > x$. Thus, for every integer x, there exists an integer y such that $y > x$. Therefore, the statement is true. □

1.E.10. Let X be a set and let $p(x)$ be a predicate. Find a logical formula representing the statement 'there are exactly two elements $x \in X$ such that $p(x)$ is true'. Use the structure of this logical formula to describe how a proof should be structured, and use this structure to prove that there are exactly two real numbers x such that $x^2 = 1$.

❖ **Solution**

$$\exists x_1, x_2 \in X\,(x_1 \neq x_2 \wedge p(x_1) \wedge p(x_2) \wedge \forall x' \in X(p(x') \Rightarrow (x' = x_1 \vee x' = x_2)))$$

This formula says: there exist two distinct elements x_1 and x_2 both satisfying p, and every element that satisfies p must be equal to one of these two.

To prove that **exactly two** elements of a set satisfy a property p, we must show two things:

- Existence of at least two distinct elements:
  Find two specific elements x_1 and x_2 (with $x_1 \neq x_2$) such that $p(x_1)$ and $p(x_2)$ both hold.
- Existence of at most two distinct elements:
  Prove that if any element x' satisfies p, then necessarily $x' = x_1$ or $x' = x_2$.

Proof
We want to prove that there are exactly two real numbers x such that $x^2 = 1$.

Let $p(x)$ be the statement $x^2 = 1$, where $x \in \mathbb{R}$.

- Existence of at least two distinct elements:
  Let $x_1 = 1$ and $x_2 = -1$. It's clear that $x_1 \neq x_2$ and $x_1^2 = 1^2 = 1$ and $x_2^2 = (-1)^2 = 1$. Hence we have two distinct elements satisfying the property.
- Existence of at most two distinct elements:
  Let x' be any real number such that $x'^2 = 1$. Then $x'^2 - 1 = 0$, i.e., $(x'-1)(x'+1) = 0$. Since the product of real numbers is zero if and only if at least one factor is zero, we have $x' - 1 = 0$ or $x' + 1 = 0$. Thus $x' = 1$ or $x' = -1$. So every element satisfying p is either 1 or -1.

Therefore, there are exactly two real numbers x such that $x^2 = 1$. □

1.E.11. Prove that

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$$

How might this logical equivalence help you to prove statements of the form 'p if and only if q'?

preview

❖ **Solution**

Proof

$$\begin{aligned} p \Leftrightarrow q &\equiv (p \Rightarrow q) \wedge (q \Rightarrow p) \\ &\equiv (p \Rightarrow q) \wedge (\neg q \vee p) \\ &\equiv (p \Rightarrow q) \wedge (\neg(\neg p) \vee \neg q) \\ &\equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q)) \end{aligned}$$

In logic, the original proposition "if p then q" is logically equivalent to its contrapositive "if not q then not p".

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$$

This logical equivalence introduces proof by contradiction. Sometimes, when the original proposition $p \Rightarrow q$ is difficult to prove, we can use proof by contradiction to prove its contrapositive $\neg q \Rightarrow \neg p$.

1.E.12. Prove using truth tables that $p \Rightarrow q \not\equiv q \Rightarrow p$. Give an example of propositions p and q such that $p \Rightarrow q$ is true but $q \Rightarrow p$ is false.

❖ **Solution**

p	q	$p \Rightarrow q$	$q \Rightarrow p$
✓	✓	✓	✓
✓	×	×	✓
×	✓	✓	×
×	×	✓	✓

Accouding to the true value table above, we can find that $p \Rightarrow q$ and $q \Rightarrow p$ have different true value. Therefore, $p \Rightarrow q \not\equiv q \Rightarrow p$.

For example, let p be "n is a rational number", let q be "n is a real number". Then we can say $p \Rightarrow q$, but we cannot say $q \Rightarrow p$.

In Questions 1.E.13 to 1.E.16, find a logical formula whose column in a truth table is as shown.

1.E.13.

p	q	
✓	✓	×
✓	×	✓
×	✓	✓
×	×	×

❖ **Solution**

This is the truth table of **exclusive OR**. The logical formula is $p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$.

1.E.14.

p	q	
✓	✓	✓
✓	×	×
×	✓	✓
×	×	×

preview

❖ **Solution**
The output is exactly the truth value of q, regardless of p, so the logical formula is just q.

1.E.15.

p	q	r	
✓	✓	✓	✓
✓	✓	×	✓
✓	×	✓	×
✓	×	×	×
×	✓	✓	×
×	✓	×	×
×	×	✓	✓
×	×	×	✓

❖ **Solution**
The output is exactly the truth value of $p \Rightarrow q$, regardless of r, so the logical formula is just $p \Rightarrow q$.

1.E.16.

p	q	r	
✓	✓	✓	✓
✓	✓	×	×
✓	×	✓	×
✓	×	×	×
×	✓	✓	✓
×	✓	×	×
×	×	✓	✓
×	×	×	×

❖ **Solution**
$r \wedge (p \Rightarrow q)$

1.E.17. A new logical operator $\uparrow$ is defined by the following rules:

(i) If a contradiction can be derived from the assumption that p is true, then $p \uparrow q$ is true;

(ii) If a contradiction can be derived from the assumption that q is true, then $p \uparrow q$ is true;

(iii) If r is any proposition, and if $p \uparrow q$, p and q are all true, then r is true.

This question explores this curious new logical operator.

(a) Prove that $p \uparrow p \equiv \neg p$, and deduce that $((p \uparrow p) \uparrow (p \uparrow p)) \equiv p$.

(b) Prove that $p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q)$ and $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$.

(c) Find a propositional formula using only the logical operator $\uparrow$ that is equivalent to $p \Rightarrow q$.

❖ **Solution**
According to the definition of $\uparrow$, we can derive the following truth table for $p \uparrow q$:

p	q	$p \uparrow q$
✓	✓	×
✓	×	✓
×	✓	✓
×	×	✓

When p and q are both false, $p \uparrow q$ is true. This shares the same idea with $\Rightarrow$. If the assumption is false, by the principle of explosion, any consequence may be derived. So under a false hypothesis, we cannot say the overall statement is false. By the LEM, it must be true.

(a) Prove that $p \uparrow p \equiv \neg p$, and deduce that $((p \uparrow p) \uparrow (p \uparrow p)) \equiv p$.

Proof
We will prove that $p \uparrow p \equiv \neg p$ by truth table.

p	p	$p \uparrow p$	$\neg p$
✓	✓	×	×
×	×	✓	✓

□

According to the truth table above, we can find that $p \uparrow p$ and $\neg p$ have the same truth value. Therefore, $p \uparrow p \equiv \neg p$. Then we have

$$((p \uparrow p) \uparrow (p \uparrow p)) \equiv (\neg p) \uparrow (\neg p) \equiv \neg(\neg p) \equiv p$$

(b) Prove that $p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q)$ and $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$.

Proof
We will prove $p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q)$ by truth table.

p	q	$p \vee q$	$(p \uparrow p)$	$(q \uparrow q)$	$(p \uparrow p) \uparrow (q \uparrow q)$
✓	✓	✓	×	×	✓
✓	×	✓	×	✓	✓
×	✓	✓	✓	×	✓
×	×	×	✓	✓	×

According to the truth table above, we can find that $p \vee q$ and $(p \uparrow p) \uparrow (q \uparrow q)$ have the same truth value. Therefore, $p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q)$.

We will prove $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$ by truth table.

p	q	$p \wedge q$	$(p \uparrow q)$	$(p \uparrow q) \uparrow (p \uparrow q)$
✓	✓	✓	×	✓
✓	×	×	✓	×
×	✓	×	✓	×
×	×	×	✓	×

According to the truth table above, we can find that $p \wedge q$ and $(p \uparrow q) \uparrow (p \uparrow q)$ have the same truth value. Therefore, $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$. □

(c) Find a propositional formula using only the logical operator $\uparrow$ that is equivalent to $p \Rightarrow q$.

preview

Proof
Since $p \Rightarrow q \equiv \neg p \vee q$, we can rewrite it as

$$\begin{aligned} p \Rightarrow q &\equiv \neg p \vee q \\ &\equiv (\neg p \uparrow \neg p) \uparrow (q \uparrow q) \\ &\equiv p \uparrow (\neg q) \end{aligned}$$

We can verify that $p \Rightarrow q \equiv p \uparrow (\neg q)$ by truth table.

p	q	$p \Rightarrow q$	$\neg q$	$p \uparrow (\neg q)$
✓	✓	✓	×	✓
✓	×	×	✓	×
×	✓	✓	×	✓
×	×	✓	✓	✓

According to the truth table above, we can find that $p \Rightarrow q$ and $p \uparrow (\neg q)$ have the same truth value. Therefore, $p \Rightarrow q \equiv p \uparrow (\neg q)$. □

1.E.18. Let X be $\mathbb{Z}$ or $\mathbb{Q}$, and define a logical formula p by:

$$\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])$$

Write out $\neg p$ as a maximally negated logical formula. Prove that p is true when $X = \mathbb{Z}$, and p is false when $X = \mathbb{Q}$.

❖ **Solution**
The maximally negated logical formula of $\neg p$ is

$$\begin{aligned} \neg p &\equiv \neg[\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])] \\ &\equiv \exists x \in X, \neg[\exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])] \\ &\equiv \exists x \in X, \forall y \in X, \neg(x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)]) \\ &\equiv \exists x \in X, \forall y \in X, [\neg(x < y) \vee \neg[\forall z \in X, \neg(x < z \wedge z < y)]] \\ &\equiv \exists x \in X, \forall y \in X, [x \geqslant y \vee \exists z \in X, (x < z \wedge z < y)] \end{aligned}$$

Prove that p is true when $X = \mathbb{Z}$

Proof
When $X = \mathbb{Z}$, we want to prove p is true. That is for every integer x, there exists an integer y such that $x < y$ and there is no integer strictly between x and y.

Let $x \in \mathbb{Z}$ be arbitrary. Let $y = x + 1$. Then $y \in \mathbb{Z}$ and $x < y$. Moreover, assume that there exists $z \in \mathbb{Z}$ and $x < z < y$, then z must be equal to $x + 1$, which is a contradiction with $z < y$. Therefore, for every integer x, there exists an integer y such that $x < y$ and there is no integer strictly between x and y. Thus, p is true when $X = \mathbb{Z}$. □

Prove that p is false when $X = \mathbb{Q}$

preview

Proof
When $X = \mathbb{Q}$, we want to prove p is false, i.e. $\neg p$ is true. That is there exists a rational number x such that for every rational number y, either $x \geqslant y$ or there exists a rational number z such that $x < z < y$.

Let $x = 0$. Let $y \in \mathbb{Q}$ be arbitrary. There are 2 cases:

- If $y \leqslant 0$, then $x \geqslant y$.
- If $y > 0$, then there exists a rational number $z = \dfrac{y}{2}$ such that $x < z < y$.

In summary, for every rational number y, either $x \geqslant y$ or there exists a rational number z such that $x < z < y$. Thus, $\neg p$ is true when $X = \mathbb{Q}$. Therefore, p is false when $X = \mathbb{Z}$. □

1.E.19. Use Definition 1.2.26 to write out a maximally negated logical formula that is equivalent to $\neg\exists! x \in X,\, p(x)$. Describe the strategy that this equivalence suggests for proving that there is not a unique $x \in X$ such that $p(x)$ is true, and use this strategy to prove that, for all $a \in \mathbb{R}$, if $a \neq -1$ then there is not a unique $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$.

❖ **Solution**

$$\begin{aligned}\neg\exists! x \in X,\, p(x) &\equiv \neg[(\exists x \in X,\, p(x)) \wedge (\forall a \in X, \forall b \in X,\, [(p(a) = p(b)) \Rightarrow a = b])] \\ &\equiv [\neg(\exists x \in X,\, p(x))] \vee [\neg(\forall a \in X, \forall b \in X,\, [(p(a) = p(b)) \Rightarrow a = b])] \\ &\equiv \underbrace{(\forall x \in X,\, \neg p(x))}_{non-existence} \vee \underbrace{(\exists a \in X, \exists b \in X,\, [(p(a) = p(b)) \wedge a \neq b])}_{non-uniqueness}\end{aligned}$$

If we want to prove that there is not a unique $x \in X$ such that $p(x)$ is true, we can show that either there does not exist any $x \in X$ such that $p(x)$ is true, or there exist at least two distinct elements a and b in X such that $p(a)$ and $p(b)$ are both true.

preview

Proof
Let $a \in \mathbb{R}$ be arbitrary and assume that $a \neq -1$. We want to prove that there is not a unique $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$.

Notice that $x^4 - 2ax^2 + a^2 - 1$ can be factored into

$$x^4 - 2ax^2 + a^2 - 1 = (x^2 - a)^2 - 1 = (x^2 - a - 1)(x^2 - a + 1)$$

Thus $x^4 - 2ax^2 + a^2 - 1 = 0$ has four roots

$$x = \pm\sqrt{a+1} \quad \text{and} \quad x = \pm\sqrt{a-1}$$

- If $a < -1$, then $a + 1 < 0$ and $a - 1 < 0$, so both roots are imaginary. Thus, there does not exist any $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$.
- If $a > -1$, then $a + 1 > 0$, so there exist at least two real roots $\pm\sqrt{a+1}$.

In summary, if $a < -1$, then there does not exist any $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$. If $a > -1$, then there exist at least two distinct elements $\pm\sqrt{a+1} \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$. Therefore, for all $a \in \mathbb{R}$, if $a \neq -1$ then there is not a unique $x \in \mathbb{R}$ such that $x^4 - 2ax^2 + a^2 - 1 = 0$. □

1.E.20. Define a new quantifier $\forall!$ such that de Morgan's laws for quantifiers (Theorem 1.3.29) hold with $\forall$ and $\exists$ replaced by $\forall!$ and $\exists!$, respectively.

❖ **Solution**
de Morgan's laws for quantifiers hold with $\forall$ and $\exists$ replaced by $\forall!$ and $\exists!$, means the following equivalences hold:

$$\neg\forall! x \in X,\, p(x) \equiv \exists! x \in X,\, \neg p(x)$$
$$\neg\exists! x \in X,\, p(x) \equiv \forall! x \in X,\, \neg p(x)$$

Thus, $\forall!$ can be defined as follows:

$$\forall! x \in X,\, p(x) \equiv \neg\exists! x \in X,\, \neg p(x)$$

We can maximally negate the right-hand side to get an equivalent definition of $\forall!$:

$$\begin{aligned}
\forall! x \in X,\, p(x) &\equiv \neg\exists! x \in X,\, \neg p(x) \\
&\equiv \neg[(\exists x \in X,\, \neg p(x)) \wedge (\forall a \in X, \forall b \in X,\, [(\neg p(a) = \neg p(b)) \Rightarrow a = b])] \\
&\equiv [\neg(\exists x \in X,\, \neg p(x))] \vee [\neg(\forall a \in X, \forall b \in X,\, [(\neg p(a) = \neg p(b)) \Rightarrow a = b])] \\
&\equiv (\forall x \in X,\, p(x)) \vee (\exists a \in X, \exists b \in X,\, [(\neg p(a) = \neg p(b)) \wedge a \neq b])
\end{aligned}$$

In words: "For all $x \in X$, $p(x)$ holds, or there are at least two distinct elements $a, b \in X$ for which $\neg p(a)$ and $\neg p(b)$ are both true" (i.e., it is not the case that exactly one element fails to satisfy p).

preview

True–False questions

In Questions 1.E.21 to 1.E.30, determine (with proof) whether the statement is true or false.

1.E.21. Every implication is logically equivalent to its contrapositive.

❖ **Solution**
True

We can verify that $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$ by truth tables.

p	q	$p \Rightarrow q$	$\neg p$	$\neg q$	$\neg q \Rightarrow \neg p$
✓	✓	✓	×	×	✓
✓	×	×	×	✓	×
×	✓	✓	✓	×	✓
×	×	✓	✓	✓	✓

According to the truth table above, we can find that $p \Rightarrow q$ and $\neg q \Rightarrow \neg p$ have the same truth value. Therefore, $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$.

1.E.22. Every implication is logically equivalent to its converse.

❖ **Solution**
False

We can verify that $p \Rightarrow q \not\equiv q \Rightarrow p$ by truth tables.

p	q	$p \Rightarrow q$	$q \Rightarrow p$
✓	✓	✓	✓
✓	×	×	✓
×	✓	✓	×
×	×	✓	✓

According to the truth table above, we can find that $p \Rightarrow q$ and $q \Rightarrow p$ have different truth values. Therefore, $p \Rightarrow q \not\equiv q \Rightarrow p$.

1.E.23. Every propositional formula whose only logical operators are conjunctions and negations is logically equivalent to a propositional formula whose only logical operators are disjunctions and negations.

❖ **Solution**
True

This is a consequence of De Morgan's laws. Any propositional formula involving only conjunctions and negations can be transformed into an equivalent formula involving only disjunctions and negations, and vice versa. For example, the formula $p \wedge q$ can be rewritten as $\neg(\neg p \vee \neg q)$, and the formula $p \vee q$ can be rewritten as $\neg(\neg p \wedge \neg q)$.

preview

1.E.24. Every propositional formula whose only logical operators are conjunctions is logically equivalent to a propositional formula whose only logical operators are disjunctions.

❖ **Solution**

False

For example, the formula $p \wedge q$ cannot be rewritten as an equivalent formula involving only disjunctions. If we try to rewrite $p \wedge q$ using only disjunctions, we would need to use negations as well, which is not allowed in this case. Therefore, not every propositional formula whose only logical operators are conjunctions is logically equivalent to a propositional formula whose only logical operators are disjunctions.

1.E.25. The formulae $p \wedge (q \vee r)$ and $(p \wedge q) \vee r$ are logically equivalent.

❖ **Solution**

False

We can verify that $p \wedge (q \vee r) \not\equiv (p \wedge q) \vee r$ by truth tables.

p	q	r	$(q \vee r)$	$p \wedge (q \vee r)$	$(p \wedge q)$	$(p \wedge q) \vee r$
✓	✓	✓	✓	✓	✓	✓
✓	✓	×	✓	✓	✓	✓
✓	×	✓	✓	✓	×	✓
✓	×	×	×	×	×	×
×	✓	✓	✓	×	×	✓
×	✓	×	✓	×	×	×
×	×	✓	✓	×	×	✓
×	×	×	×	×	×	×

According to the truth table above, we can find that $p \wedge (q \vee r)$ and $(p \wedge q) \vee r$ have different truth values. Therefore, $p \wedge (q \vee r) \not\equiv (p \wedge q) \vee r$.

1.E.26. The formulae $p \vee (q \vee r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.

❖ **Solution**

True

We can verify that $p \vee (q \vee r) \equiv (p \vee q) \wedge (p \vee r)$ by truth tables.

p	q	r	$(q \vee r)$	$p \vee (q \vee r)$	$(p \vee q)$	$(p \vee r)$	$(p \vee q) \wedge (p \vee r)$
✓	✓	✓	✓	✓	✓	✓	✓
✓	✓	×	✓	✓	✓	✓	✓
✓	×	✓	✓	✓	✓	✓	✓
✓	×	×	×	✓	✓	✓	✓
×	✓	✓	✓	✓	✓	✓	✓
×	✓	×	✓	✓	✓	×	✓
×	×	✓	✓	✓	×	✓	✓
×	×	×	×	×	×	×	×

According to the truth table above, we can find that $p \vee (q \vee r)$ and $(p \vee q) \wedge (p \vee r)$ have the same truth values. Therefore, $p \vee (q \vee r) \equiv (p \vee q) \wedge (p \vee r)$.

1.E.27. The logical formulae $\neg\forall x \in X, \forall y \in Y, p(x,y)$ and $\exists x \in X, \forall y \in Y, p(x,y)$ are logically equivalent.

❖ **Solution**
False

Proof
The maximally negated version of the original logical formula is:

$$\neg\forall x \in X, \forall y \in Y, p(x,y) \equiv \exists x \in X, \exists y \in Y, \neg p(x,y)$$

It is not the same as the given logical formula. Therefore,

$$\neg\forall x \in X, \forall y \in Y, p(x,y) \not\equiv \exists x \in X, \forall y \in Y, p(x,y)$$

1.E.28. $\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\forall x < 0, \exists y \notin \mathbb{R}, y^2 \neq x$.

❖ **Solution**
False

Proof
The maximally negated version of the original logical formula is:

$$\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x \equiv \exists x \geqslant 0, \forall y \in \mathbb{R}, y^2 \neq x$$

It is not the same as the given logical formula. Therefore,

$$\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x \not\equiv \forall x < 0, \exists y \notin \mathbb{R}, y^2 \neq x$$

1.E.29. $\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\exists x \geqslant 0, \forall y \in \mathbb{R}, y^2 \neq x$.

❖ **Solution**
True

Proof
The maximally negated version of the original logical formula is:

$$\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x \equiv \exists x \geqslant 0, \forall y \in \mathbb{R}, y^2 \neq x$$

Obviously, the given logical formula is the same as the maximally negated version of the original formula, so the they are equivalent. □

1.E.30. $\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x$ is logically equivalent to $\exists x < 0, \forall y \in \mathbb{R}, y^2 = x$.

preview

❖ **Solution**
False

Proof
The maximally negated version of the original logical formula is:

$$\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x \equiv \exists x \geqslant 0, \forall y \in \mathbb{R}, y^2 \neq x$$

In this problem, the discussion scope of the given logical formula has changed, so it is necessarily not equivalent to the original proposition. Therefore,

$$\neg\forall x \geqslant 0, \exists y \in \mathbb{R}, y^2 = x \not\equiv \exists x < 0, \forall y \in \mathbb{R}, y^2 = x$$

Always–Sometimes–Never questions

In Questions 1.E.31 to 1.E.35, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

1.E.31. Let p and q be propositions and assume that p is true. Then $p \Rightarrow q$ is true.

❖ **Solution**
Sometimes

By the definition of implication, assume that p is true

- If q is true, then $p \Rightarrow q$ is true.
- If q is false, then $p \Rightarrow q$ is false.

1.E.32. Let p and q be propositions and assume that p is false. Then $p \Rightarrow q$ is true.

❖ **Solution**
Always

By the definition of implication, assume that p is false, then $p \Rightarrow q$ is true regardless of the truth value of q.

1.E.33. Let X and Y be sets and let $p(x,y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \forall y \in Y, p(x,y)$ and $\forall y \in Y, \forall x \in X, p(x,y)$ are logically equivalent.

❖ **Solution**
Always

The two statements $\forall x \in X, \forall y \in Y, p(x,y)$ and $\forall y \in Y, \forall x \in X, p(x,y)$ are always logically equivalent because both assert that $p(x,y)$ holds for every ordered pair $(x,y) \in X \times Y$. More formally:

- If $\forall x \in X, \forall y \in Y, p(x,y)$ is true, then for every $x \in X$ and every $y \in Y$, $p(x,y)$ holds. In particular, for every $y \in Y$ and every $x \in X$, $p(x,y)$ holds, so $\forall y \in Y, \forall x \in X, p(x,y)$ is true.

- Conversely, if $\forall y \in Y, \forall x \in X, p(x,y)$ is true, then for every $y \in Y$ and every $x \in X$, $p(x,y)$ holds. In particular, for every $x \in X$ and every $y \in Y$, $p(x,y)$ holds, so $\forall x \in X, \forall y \in Y, p(x,y)$ is true.

1.E.34. Let X and Y be sets and let $p(x,y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \exists y \in Y, p(x,y)$ and $\exists y \in Y, \forall x \in X, p(x,y)$ are logically equivalent.

❖ **Solution**

Sometimes

- **Case 1: A case where the statements are equivalent**

  Let X, Y be the integer set and let $p(x,y)$ be the predicate "x divides y". Since every integer divides 0, let $y = 0$, then $\forall x \in X, \exists y \in Y, p(x,y)$ is true, and $\exists y \in Y, \forall x \in X, p(x,y)$ is also true. Therefore, the two statements are logically equivalent in this case.

- **Case 2: A case where the statements are not equivalent**

  Let X be the set of odd integers, Y be the integer set, and let $p(x,y)$ be the predicate "$x = 2y+1$". Since every odd integer can be written as $2y+1$ for some integer y, $\forall x \in X, \exists y \in Y, p(x,y)$ is true. However, there does not exist an integer y such that $2y+1$ represents all odd integers, so $\exists y \in Y, \forall x \in X, p(x,y)$ is false. Therefore, the two statements are not logically equivalent in this case.

In fact, $\exists y \in Y, \forall x \in X, p(x,y)$ is stronger than $\forall x \in X, \exists y \in Y, p(x,y)$. $\exists y \in Y, \forall x \in X, p(x,y)$ always implies $\forall x \in X, \exists y \in Y, p(x,y)$. So if $\exists y \in Y, \forall x \in X, p(x,y)$ is true, then $\forall x \in X, \exists y \in Y, p(x,y)$ must be true. However, the converse does not hold. Therefore, the two statements are sometimes logically equivalent.

1.E.35. Let X and Y be sets and let $p(x,y)$ be a predicate with free variables $x \in X$ and $y \in Y$. Then the logical formulae $\forall x \in X, \forall y \in Y, p(x,y)$ and $\exists x \in X, \exists y \in Y, \neg p(x,y)$ are logically equivalent.

❖ **Solution**

Never

$$\exists x \in X, \exists y \in Y, \neg p(x,y) \equiv \neg(\forall x \in X, \forall y \in Y, p(x,y))$$

which is not the same as $\forall x \in X, \forall y \in Y, p(x,y)$. In fact, $\exists x \in X, \exists y \in Y, \neg p(x,y)$ is the negation of $\forall x \in X, \forall y \in Y, p(x,y)$. Therefore, the two statements are never logically equivalent.

preview

Chapter 2

Sets

preview

Section 2.E

Chapter 2 exercises

Sets

2.E.1. Express the following sets in the indicated form of notation.

(a) $\{n \in \mathbb{Z} \mid n^2 < 20\}$ in list notation;

(b) $\{4k+3 \mid k \in \mathbb{N}\}$ in implied list notation;

(c) The set of all odd multiples of six in set-builder notation;

(d) The set $\{1,2,5,10,17,\dots,n^2+1,\dots\}$ in set-builder notation.

❖ **Solution**

(a) $\{-4,-3,-2,-1,0,1,2,3,4\}$;

(b) $\{3,7,11,15,\dots\}$;

(c) $\{6\cdot(2k+1) \mid k \in \mathbb{Z}\}$;

(d) $\{n^2+1 \mid n \in \mathbb{N}\}$.

2.E.2. Find sets X_n for each $n \in \mathbb{N}$ such that $X_{n+1} \subsetneqq X_n$ for all $n \in \mathbb{N}$. Can any of the sets X_n be empty?

❖ **Solution**

Let $n \in \mathbb{N}$, define $X_n = \{k \in \mathbb{N} \mid k \leqslant n\}$. Then

$$X_{n+1} = \{n+1, n+2, \dots\} \subsetneqq \{n, n+1, n+2, \dots\} = X_n$$

Thus $X_{n+1} \subsetneqq X_n$ for every $n \in \mathbb{N}$.

Can any of the sets X_n be empty? **NO**!

Assume X_m is empty for some m, then we have

$$X_{m+1} \subsetneqq X_m = \varnothing$$

but $\varnothing$ has no proper subset. Therefore there is no sets X_n is empty.

2.E.3. Express the set $\mathscr{P}(\{\varnothing, \{\varnothing, \{\varnothing\}\}\})$ in list notation.

❖ **Solution**

The set $\{\varnothing, \{\varnothing, \{\varnothing\}\}\}$ has two elements, which are $\varnothing$ and $\{\varnothing, \{\varnothing\}\}$, so the power set is

$$\mathscr{P}(\{\varnothing, \{\varnothing, \{\varnothing\}\}\}) = \{\varnothing, \{\varnothing\}, \{\{\varnothing, \{\varnothing\}\}\}, \{\varnothing, \{\varnothing, \{\varnothing\}\}\}\}$$

preview

2.E.4. Let X be a set and let $U, V \subseteq X$. Prove that U and V are disjoint if and only if $U \subseteq X \setminus V$.

❖ **Proof**

Let X be a set and let $U, V \subseteq X$. We want to prove

$$U \cap V = \varnothing \iff U \subseteq X \setminus V$$

- ($\Longrightarrow$) If $U \cap V = \varnothing$, then for any $x \in U$, we have $x \notin V$. Since $U \subseteq X$, we have $x \in X$. Since $x \in X$ and $x \notin V$, we get $x \in X \setminus V$. Since this holds for every $x \in U$, therefore $U \subseteq X \setminus V$.
- ($\Longleftarrow$) If $U \subseteq X \setminus V$, then for any $x \in U$, we have $x \in X \setminus V$, thus $x \notin V$. Therefore $U \cap V = \varnothing$.

In conclusion, $U \cap V = \varnothing \iff U \subseteq X \setminus V$. □

2.E.5. For each of the following statements, determine whether or not it is true for all sets A and X, and prove your claim.

(a) If $X \setminus A = \varnothing$, then $X = A$.

(b) If $X \setminus A = X$, then $A = \varnothing$.

(c) If $X \setminus A = A$, then $A = \varnothing$.

(d) $X \setminus (X \setminus A) = A$.

❖ **Solution**

(a) **False**.
Here is a counterexample: let $X = \{1\}$ and $A = \{1,2\}$. Then $X \setminus A = \varnothing$, but $X \neq A$.

(b) **False**.
Here is a counterexample: let $X = \{1\}$ and $A = \{2\}$. Then $X \setminus A = \{1\} = X$, but $X \neq \varnothing$.

(c) **True**.
Assume $A \neq \varnothing$. Then A has at least one element $a \in A$. Since $A = X \setminus A$, we have $a \in X \setminus A$. Thus $a \notin A$, contradicting $a \in A$. Therefore A must be empty.

(d) **False**.
Here is a counterexample: let $X = \{1\}$ and $A = \{2\}$. Then $X \setminus A = \{1\}$, $X \setminus (X \setminus A) = \varnothing$, but $A \neq \varnothing$.

2.E.6. For each of the following statements, determine whether it is true for all sets X, Y, false for all sets X, Y, or true for some choices of X and Y and false for others.

(a) $\mathscr{P}(X \cup Y) = \mathscr{P}(X) \cup \mathscr{P}(Y)$

(b) $\mathscr{P}(X \cap Y) = \mathscr{P}(X) \cap \mathscr{P}(Y)$

(c) $\mathscr{P}(X \times Y) = \mathscr{P}(X) \times \mathscr{P}(Y)$

(d) $\mathscr{P}(X \setminus Y) = \mathscr{P}(X) \setminus \mathscr{P}(Y)$

❖ **Solution**

(a) **Not true for all sets, but true for some choices.**
Here is a counterexample: Let $X = \{1\}, Y = \{2\}$, then $X \cup Y = \{1,2\} \in \mathscr{P}(X \cup Y)$.
But $\mathscr{P}(X) \cup \mathscr{P}(Y) = \{\varnothing, \{1\}, \{2\}\}$ and $\{1,2\} \notin \mathscr{P}(X) \cup \mathscr{P}(Y)$.
Therefore, $\mathscr{P}(X \cup Y) \neq \mathscr{P}(X) \cup \mathscr{P}(Y)$ for all sets X, Y.
However, if $X \subseteq Y$, then $X \cup Y = Y$. Thus $\mathscr{P}(X \cup Y) = \mathscr{P}(Y) = \mathscr{P}(X) \cup \mathscr{P}(Y)$.

(b) **True for all sets.**
Assume set $S \in \mathscr{P}(X \cap Y)$, then $S \subseteq X \cap Y$. Thus $S \subseteq X$ and $S \subseteq Y$. Hence $S \in \mathscr{P}(X)$ and $S \in \mathscr{P}(Y)$. Hence $S \in \mathscr{P}(X) \cap \mathscr{P}(Y)$.

Therefore,

$$\mathscr{P}(X \cap Y) = \mathscr{P}(X) \cap \mathscr{P}(Y)$$

for all sets X, Y.

(c) **False for all sets.**
For any sets X, Y, since the empty set is a subset of every set, we have $\varnothing \in \mathscr{P}(X \times Y)$.

But every element of $\mathscr{P}(X) \times \mathscr{P}(Y)$ is an ordered pair (A, B). Hence $\varnothing \notin \mathscr{P}(X) \times \mathscr{P}(Y)$.

Therefore,

$$\mathscr{P}(X \times Y) \neq \mathscr{P}(X) \times \mathscr{P}(Y)$$

for all sets X, Y.

(d) **False for all sets.**
For any sets X, Y, since the empty set is a subset of every set, we have $\varnothing \in \mathscr{P}(X \setminus Y)$.

we also have $\varnothing \in \mathscr{P}(X)$ and $\varnothing \in \mathscr{P}(Y)$. Hence $\varnothing \notin \mathscr{P}(X) \setminus \mathscr{P}(Y)$.

Therefore,

$$\mathscr{P}(X \setminus Y) \neq \mathscr{P}(X) \setminus \mathscr{P}(Y)$$

for all sets X, Y.

2.E.7. Let F be a set whose elements are all sets. Prove that if $\forall A \in F,\ \forall x \in A,\ x \in F$, then $F \subseteq \mathscr{P}(F)$.

❖ **Proof**
Take any set $A \in F$. By the hypothesis, for every $x \in A$, we have $x \in F$. By definition of the subset,

$$A \subseteq F$$

By definition of power set,

$$A \in \mathscr{P}(F)$$

Since $A \in F$ is arbitrary, every element of F is an element of $\mathscr{P}(F)$. Hence

$$F \subseteq \mathscr{P}(F)$$

This proves the claim. □

Questions 2.E.9 to 2.E.14 concern the *symmetric difference* of sets, defined below.

✦ **Definition 2.E.8**
The **symmetric difference** of sets X and Y is the set $X \triangle Y$ (LaTeX code: `\triangle`) defined by

$$X \triangle Y = \{a \mid a \in X \text{ or } a \in Y \text{ but not both}\}$$

2.E.9. Prove that $X \triangle Y = (X \setminus Y) \cup (Y \setminus X) = (X \cup Y) \setminus (X \cap Y)$ for all sets X and Y.

preview

❖ **Proof**

First, we prove that $X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$ by double containment.

- ($\subseteq$) By the definition of **symmetric difference**, if $a \in X \triangle Y$, then there are two cases:
  - ◇ If $a \in X$, then $a \notin Y$, thus $a \in X \setminus Y$.
  - ◇ If $a \in Y$, then $a \notin X$, thus $a \in Y \setminus X$.

  Hence $a \in (X \setminus Y) \cup (Y \setminus X)$.

  Since element a is arbitrary, we shown that for any $a \in X \triangle Y$, $a \in (X \setminus Y) \cup (Y \setminus X)$.

  Therefore,
  $$X \triangle Y \subseteq (X \setminus Y) \cup (Y \setminus X)$$
- ($\supseteq$) If $a \in (X \setminus Y) \cup (Y \setminus X)$, then there are two cases:
  - ◇ If $a \in X \setminus Y$, then $a \in X$ and $a \notin Y$. By the definition, we have $a \in X \triangle Y$.
  - ◇ If $a \in Y \setminus X$, then $a \in Y$ and $a \notin X$. By the definition, we have $a \in X \triangle Y$.

  Hence $a \in X \triangle Y$.

  Since element a is arbitrary, we shown that for any $a \in (X \setminus Y) \cup (Y \setminus X)$, $a \in X \triangle Y$.

  Therefore,
  $$X \triangle Y \supseteq (X \setminus Y) \cup (Y \setminus X)$$

By double containment, we show that

$$X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$$

Next, we prove that $X \triangle Y = (X \cup Y) \setminus (X \cap Y)$ by logical equivalence.

We can rewrite the definition of **symmetric difference** formally as

$$X \triangle Y = \{a \mid (a \in X \vee a \in Y) \wedge \neg(a \in X \cap Y)\}$$

Then

$$\begin{aligned} a \in X \triangle Y &\iff (a \in X \vee a \in Y) \wedge \neg(a \in X \cap Y) \\ &\iff (a \in X \cup Y) \wedge \neg(a \in X \cap Y) \\ &\iff a \in (X \cup Y) \setminus (X \cap Y) \end{aligned}$$

Therefore, $X \triangle Y = (X \cup Y) \setminus (X \cap Y)$.

In conclusion,
$$X \triangle Y = (X \setminus Y) \cup (Y \setminus X) = (X \cup Y) \setminus (X \cap Y)$$
for all sets X and Y. □

2.E.10. Let X be a set. Prove that $X \triangle X = \varnothing$ and $X \triangle \varnothing = X$.

preview

❖ **Proof**

According to the conclusion of Question 2.E.9, we have

$$X \triangle X = (X \setminus X) \cup (X \setminus X) = \varnothing \cup \varnothing = \varnothing$$

and

$$X \triangle \varnothing = (X \setminus \varnothing) \cup (\varnothing \setminus X) = X \cup \varnothing = X$$

2.E.11. Let X and Y be sets. Prove that $X = Y$ if and only if $X \triangle Y = \varnothing$.

❖ **Proof**

We want to prove $X = Y \iff X \triangle Y = \varnothing$.

- ($\Longrightarrow$) If $X = Y$, according to the conclusion of Question 2.E.10, it's clear that $X \triangle Y = \varnothing$.
- ($\Longleftarrow$) If $X \triangle Y = \varnothing$, by Question 2.E.9, we have $(X \setminus Y) \cup (Y \setminus X) = \varnothing$. Thus $X \setminus Y = \varnothing$ and $Y \setminus X = \varnothing$. Hence $X = Y$.

In conclusion, $X = Y$ if and only if $X \triangle Y = \varnothing$. □

2.E.12. Prove that sets X and Y are disjoint if and only if $X \triangle Y = X \cup Y$.

❖ **Proof**

We want to prove $X \cap Y = \varnothing \iff X \triangle Y = X \cup Y$.

- ($\Longrightarrow$) If $X \cap Y = \varnothing$, then By Question 2.E.9, we have

$$X \triangle Y = (X \cup Y) \setminus (X \cap Y) = (X \cup Y) \setminus \varnothing = X \cup Y$$

- ($\Longleftarrow$) If $X \triangle Y = X \cup Y$, then By Question 2.E.9, we have

$$X \triangle Y = (X \cup Y) \setminus (X \cap Y) = X \cup Y$$

  Hence $X \cap Y = \varnothing$.

In conclusion, sets X and Y are disjoint if and only if $X \triangle Y = X \cup Y$. □

2.E.13. Prove that $X \triangle (Y \triangle Z) = (X \triangle Y) \triangle Z$ for all sets X, Y and Z.

❖ **Proof**

We will prove this proposition by truth tables.

$a \in X$	$a \in Y$	$a \in Z$	$a \in (X \triangle Y)$	$a \in (Y \triangle Z)$	$a \in X \triangle (Y \triangle Z)$	$a \in (X \triangle Y) \triangle Z$
✓	✓	✓	×	×	✓	✓
✓	✓	×	×	✓	×	×
✓	×	✓	✓	✓	×	×
✓	×	×	✓	×	✓	✓
×	✓	✓	✓	×	×	×
×	✓	×	✓	✓	✓	✓
×	×	✓	×	✓	✓	✓
×	×	×	×	×	×	×

preview

According to the truth table above, we can find that for any a, $a \in X \triangle (Y \triangle Z)$ and $a \in (X \triangle Y) \triangle Z$ have the same truth value. Therefore, $X \triangle (Y \triangle Z) = (X \triangle Y) \triangle Z$ for all sets X, Y and Z. □

2.E.14. Prove that $X \cap (Y \triangle Z) = (X \cap Y) \triangle (X \cap Z)$ for all sets X, Y and Z.

❖ **Proof**

We will prove this proposition by truth tables.

Let set $P = X \cap (Y \triangle Z)$ and set $Q = (X \cap Y) \triangle (X \cap Z)$.

$a \in X$	$a \in Y$	$a \in Z$	$a \in (Y \triangle Z)$	$a \in (X \cap Y)$	$a \in (X \cap Z)$	$a \in P$	$a \in Q$
✓	✓	✓	×	✓	✓	×	×
✓	✓	×	✓	✓	×	✓	✓
✓	×	✓	✓	×	✓	✓	✓
✓	×	×	×	×	×	×	×
×	✓	✓	×	×	×	×	×
×	✓	×	✓	×	×	×	×
×	×	✓	✓	×	×	×	×
×	×	×	×	×	×	×	×

According to the truth table above, we can find that for any a, $a \in X \cap (Y \triangle Z)$ and $a \in (X \cap Y) \triangle (X \cap Z)$ have the same truth value. Therefore, $X \cap (Y \triangle Z) = (X \cap Y) \triangle (X \cap Z)$ for all sets X, Y and Z. □

✦ **Definition 2.E.15**

A subset $U \subseteq \mathbb{R}$ is **open** if, for all $a \in U$, there exists $\delta > 0$ such that $(a - \delta, a + \delta) \subseteq U$.

In Questions Questions 2.E.16 to 2.E.19 you will prove some elementary facts about open subsets of $\mathbb{R}$.

2.E.16. For each of the following subsets of $\mathbb{R}$, determine (with proof) whether it is open:

(a) $\varnothing$;

(b) $(0,1)$;

(c) $(0,1]$;

(d) $\mathbb{Z}$;

(e) $\mathbb{R} \setminus \mathbb{Z}$;

(f) $\mathbb{Q}$.

❖ **Solution**

(a) $\varnothing$ is open.

This conclusion may be somewhat counterintuitive. According to the definition of open set, we need to check whether every element in the set U satisfies the condition,

$$\forall a \in U, \exists \delta > 0, (a - \delta, a + \delta) \subseteq U$$

But the empty set has no elements at all! How should we examine it?

We can consider the maximally negated form of the definition:

$$\neg (\forall a \in U, \exists \delta > 0, (a - \delta, a + \delta) \subseteq U) \iff \exists a \in U, \forall \delta > 0, \neg ((a - \delta, a + \delta) \subseteq U)$$

Since the empty set contains no elements, it clearly does not satisfy the negated definition of open set. By the law of excluded middle and double negation, the empty set must be open.

(b) $(0,1)$ is open.
Let $a \in (0,1)$ be arbitrary. Choose

$$\delta = \min(1-a, a) > 0$$

Since $\delta \leqslant 1-a$, then

$$a + \delta \leqslant a + (1-a) = 1$$

Since $\delta \leqslant a$, then

$$a - \delta \geqslant a - a = 0$$

Hence,

$$(a-\delta, a+\delta) \subseteq (0,1)$$

Therefore, $(0,1)$ is open.

(c) $(0,1]$ is not open.
Consider the $a = 1 \in (0,1]$. For any $\delta > 0$, we have $1+\delta > 1$, thus

$$(1-\delta, 1+\delta) \nsubseteq (0,1]$$

Therefore, $(0,1]$ is not open.

(d) $\mathbb{Z}$ is not open.
Let $a \in \mathbb{Z}$. For any $\delta > 0$, there always exists $\varepsilon = \min\left(\frac{\delta}{2}, \frac{1}{2}\right)$, such that

$$a + \varepsilon \in (a-\delta, a+\delta)$$

Since $0 < \varepsilon < 1$, so $a + \varepsilon \notin \mathbb{Z}$. Hence

$$(a-\delta, a+\delta) \nsubseteq \mathbb{Z}$$

Therefore, $\mathbb{Z}$ is not open.

(e) $\mathbb{R} \setminus \mathbb{Z}$ is open.
$\mathbb{R} \setminus \mathbb{Z}$ means removing all integers from real numbers. It partitions the real number set into many open intervals defined by consecutive integers.

$$\mathbb{R} \setminus \mathbb{Z} = \bigcup_{p \in \mathbb{Z}} (p, p+1)$$

For any $a \in \mathbb{R} \setminus \mathbb{Z}$, a lies strictly between some consecutive integers,

$$p < a < p+1, \quad \text{where } p = \lfloor a \rfloor$$

Choose

$$\delta = \min(a-p, p+1-a) > 0$$

Since $\delta \leqslant p+1-a$, then

$$a + \delta \leqslant a + (p+1-a) = p+1$$

preview

Since $\delta \leqslant a-p$, then

$$a-\delta \geqslant a-(a-p)=p$$

Hence,

$$(a-\delta, a+\delta) \subseteq (p, p+1) \subseteq \mathbb{R}\setminus\mathbb{Z}$$

Therefore, $\mathbb{R}\setminus\mathbb{Z}$ is open.

(f) $\mathbb{Q}$ is not open.

Let $a \in \mathbb{Q}$. For any $\delta > 0$, choose an integer N large enough such that $\dfrac{\sqrt{2}}{N} < \delta$. Then

$$a+\frac{\sqrt{2}}{N} \in (a, a+\delta) \subset (a-\delta, a+\delta)$$

Since $a \in \mathbb{Q}$ but $\dfrac{\sqrt{2}}{N} \notin \mathbb{Q}$, we have $a+\dfrac{\sqrt{2}}{N} \notin \mathbb{Q}$. Hence

$$(a-\delta, a+\delta) \nsubseteq \mathbb{Q}$$

Therefore, $\mathbb{Q}$ is not open.

2.E.17. Prove that a subset $U \subseteq \mathbb{R}$ is open if and only if, for all $a \in U$, there exist $u, v \in \mathbb{R}$ such that $u < a < v$ and $(u, v) \subseteq U$.

❖ **Proof**

We want to prove

$$U \subseteq \mathbb{R} \text{ is open } \iff \forall a \in U, \exists u, v \in \mathbb{R}, (u < a < v) \wedge ((u, v) \subseteq U)$$

- ($\Longrightarrow$) This is very close to the definition of open set.

  Suppose $U \subseteq \mathbb{R}$ is open. Let $a \in U$. By the definition of open set, there exists $\delta > 0$ such that

  $$(a-\delta, a+\delta) \subseteq U$$

  Let

  $$u = a-\delta, \quad v = a+\delta$$

  Then $u < a < v$, and $(u, v) = (a-\delta, a+\delta) \subseteq U$. So the desired condition holds.

- ($\Longleftarrow$) Suppose that for every $a \in U \subseteq \mathbb{R}$, there exist $u, v \in \mathbb{R}$ such that

  $$u < a < v \quad \text{and} \quad (u, v) \subseteq U$$

  Define

  $$\delta = \min(a-u, v-a) > 0$$

  Since $\delta \leqslant v-a$, then

  $$a+\delta \leqslant a+(v-a) = v$$

  Since $\delta \leqslant a-u$, then

  $$a-\delta \geqslant a-(a-u) = u$$

  Therore,

  $$(a-\delta, a+\delta) \subseteq (u, v) \subseteq U$$

  Thus for every $a \in U \subseteq \mathbb{R}$, there exists $\delta > 0$ such that $(a-\delta, a+\delta) \subseteq U$. Hence, U is open. □

2.E.18. In this question you will prove that the intersection of finitely many open sets is open, but the intersection of infinitely many open sets might not be open.

(a) Let $n \geqslant 1$ and suppose $U_1, U_2, \ldots, U_n$ are open subsets of $\mathbb{R}$. Prove that the intersection $U_1 \cap U_2 \cap \cdots \cap U_n$ is open.

(b) Prove that $(0, 1 + \frac{1}{n})$ is open for all $n \geqslant 1$, but that $\bigcap_{n \geqslant 1} (0, 1 + \frac{1}{n})$ is not open.

❖ **Solution**

(a) Suppose $U_1, U_2, \ldots, U_n \in \mathbb{R}$ be open.

- If $U_1 \cap U_2 \cap \cdots \cap U_n = \varnothing$. By the conclusion of Question 2.E.16 part (a), $\varnothing$ is open.
- If $U_1 \cap U_2 \cap \cdots \cap U_n \neq \varnothing$. Let

$$a \in U_1 \cap U_2 \cap \cdots \cap U_n$$

Since for every $1 \leqslant i \leqslant n$, each U_i is open and $x \in U_i$, there exists $\delta_i > 0$ such that

$$(a - \delta_i, a + \delta_i) \subseteq U_i$$

Define

$$\delta = \min(\delta_1, \delta_2, \ldots, \delta_n)$$

Because this is a minimum of finitely many positive numbers, we have $\delta > 0$. Then for every $1 \leqslant i \leqslant n$,

$$(a - \delta, a + \delta) \subseteq (a - \delta_i, a + \delta_i) \subseteq U_i$$

Thus

$$(a - \delta, a + \delta) \subseteq U_1 \cap U_2 \cap \cdots \cap U_n$$

By Definition 2.E.15, $U_1 \cap U_2 \cap \cdots \cap U_n$ is open.

In conclusion, $U_1 \cap U_2 \cap \cdots \cap U_n$ is open.

(b) First, we prove $(0, 1 + \frac{1}{n})$ is open for all $n \geqslant 1$.

Let $a \in (0, 1 + \frac{1}{n})$ be arbitrary, then $0 < a < 1 + \frac{1}{n}$. Choose

$$\delta = \min\left(a, 1 + \frac{1}{n} - a\right) > 0$$

Since $\delta \leqslant a$, then

$$a - \delta \geqslant a - a \ = 0$$

Since $\delta \leqslant 1 + \frac{1}{n} - a$, then

$$a + \delta \leqslant a + (1 + \frac{1}{n} - a) = 1 + \frac{1}{n}$$

Hence,

$$(a - \delta, a + \delta) \subseteq (0, 1 + \frac{1}{n})$$

Therefore, $(0, 1 + \frac{1}{n})$ is open.

Next, we prove $\bigcap_{n \geqslant 1} (0, 1 + \frac{1}{n})$ is not open.

preview

Since $(1+\frac{1}{n})$ decreases monotonically with the increase of n, and

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)=1$$

Therefore

$$\bigcap_{n\geqslant 1}\left(0,1+\frac{1}{n}\right)=(0,1]$$

By the conclusion of Question 2.E.16 part (c), $(0,1]$ is not open.

2.E.19. Prove that a subset $U\subseteq\mathbb{R}$ is open if and only if it can be expressed as a union of open intervals—more precisely, $U\subseteq\mathbb{R}$ is open if and only if, for some indexing set I, there exist real numbers a_i,b_i for each $i\in I$, such that $U=\bigcup_{i\in I}(a_i,b_i)$.

❖ **Proof**

We want to prove

$$U\subseteq\mathbb{R}\text{ is open }\iff\exists I,\forall i\in I,\exists a_i,b_i\in\mathbb{R},U=\bigcup_{i\in I}(a_i,b_i)$$

- ($\Longrightarrow$) Suppose $U\in\mathbb{R}$ is open. For each $x\in U$, by Definition 2.E.15 there exists $\delta>0$ such that
$$(x-\delta_x,x+\delta_x)\subseteq U$$
Let index set $I=U$. For each $x\in I$, let $a_x=x-\delta$ and $b_x=x+\delta$. Obviously, $a_x,b_x\in\mathbb{R}$. Now we have
$$x\in(a_x,b_x)\subset U$$
Consider the union set
$$V=\bigcup_{x\in I}(a_x,b_x)$$
We prove $V=U$ by double containment:
  - ◇ ($V\subseteq U$) Because for each $x\in I$, $x\in(a_x,b_x)\subseteq U$, thus the union of these subset is subset of U. Hence $V\subseteq U$.
  - ◇ ($U\subseteq V$) Let $y\in U$ be arbitrary, then $y\in(y-\delta_y,y+\delta_y)\subseteq\bigcup_{y\in I}(a_y,b_y)$. Hence $U\subseteq V$

  Therefore, $U=\bigcup_{i\in I}(a_i,b_i)$
- ($\Longleftarrow$) Suppose $U=\bigcup_{i\in I}(a_i,b_i)$. Let $x\in U$ be arbitrary. By the definition of union set, there must exists $i\in I$ such that $x\in(a_i,b_i)$. Let
$$\delta=\min(x-a_i,b_i-x)>0$$
  - ◇ Since $\delta\leqslant x-a_i$, then $x-\delta\geqslant a_i$;
  - ◇ Since $\delta\leqslant b_i-x$, then $x+\delta\leqslant b_i$.

Therefore, $(x-\delta, x+\delta) \subseteq (a_i, b_i)$. Since $(a_i, b_i) \subseteq \bigcup_{i \in I} (a_i, b_i) = U$, thus $(x-\delta, x+\delta) \subset U$.

By Definition 2.E.15, U is open.

In conclusion, a subset $U \subseteq \mathbb{R}$ is open if and only if it can be expressed as a union of open intervals. □

2.E.20. Let $\{A_i \mid i \in \mathbb{N}\}$ and $\{B_j \mid j \in \mathbb{N}\}$ be families of sets such that:

(i) For all $i \in \mathbb{N}$, there exists some $j \geqslant i$ such that $B_j \subseteq A_i$; and

(ii) For all $j \in \mathbb{N}$, there exists some $i \geqslant j$ such that $A_i \subseteq B_j$.

Prove that $\bigcap_{i \in \mathbb{N}} A_i = \bigcap_{j \in \mathbb{N}} B_j$.

❖ **Proof**

We prove $\bigcap_{i \in \mathbb{N}} A_i = \bigcap_{j \in \mathbb{N}} B_j$ by double containment:

- ($\subseteq$) Let $x \in \bigcap_{i \in \mathbb{N}} A_i$ be arbitrary. By the definition of the intersection, this means that $x \in A_i$ for all $i \in \mathbb{N}$.

  Let $j \in \mathbb{N}$ be given. By condition (ii), there exists some integer $i \geqslant j$ such that:

  $$A_i \subseteq B_j$$

  Since x is an element of every set in the family $\{A_i\}$, so $x \in B_j$.

  Since $j \in \mathbb{N}$ is arbitrary, so $x \in B_j$ holds for all $j \in \mathbb{N}$. Therefore:

  $$x \in \bigcap_{j \in \mathbb{N}} B_j$$

  This prove that $\bigcap_{i \in \mathbb{N}} A_i \subseteq \bigcap_{j \in \mathbb{N}} B_j$

- ($\supseteq$) The proof is entirely symmetric. We can exchange A and B, using the same argument as above, to prove that $\bigcap_{j \in \mathbb{N}} B_j \subseteq \bigcap_{i \in \mathbb{N}} A_i$

By double containment, we have $\bigcap_{i \in \mathbb{N}} A_i = \bigcap_{j \in \mathbb{N}} B_j$. □

True–False questions

In Questions 2.E.21 to 2.E.26, determine (with proof) whether the statement is true or false.

2.E.21. If E is a set and $\neg \forall x, x \in E$, then $E = \varnothing$.

preview

❖ **Solution**
False

$\neg\forall x, x \in E$ means not every x belongs to E, i.e. there is at least one object not in E. That does not imply E has no elements.

For example, let $E = \{\varnothing\}$, then $E \neq \varnothing$. However, not every set x is an element if E, i.e. $x = \mathbb{N} \notin E$.

2.E.22. If E is a set and $\forall x, \neg x \in E$, then $E = \varnothing$.

❖ **Solution**
True

$\forall x, \neg x \in E$ means all of x does not in E, i.e. E has no elements at all. By the defination of empty set, $E = \varnothing$.

2.E.23. $\{1,2,3\} = \{1,2,1,3,2,1\}$

❖ **Solution**
True

Sets ignore repetition and order, so $\{1,2,3\}$ and $\{1,2,1,3,2,1\}$ have the same elements, which are $1,2,3$.

2.E.24. $\varnothing \in \varnothing$

❖ **Solution**
False

By the defination of empty set, there is no element in the empty set.

2.E.25. $\varnothing \in \{\varnothing\}$

❖ **Solution**
True

Set $\{\varnothing\}$ has exactly one element, which is $\varnothing$.

2.E.26. $\varnothing \in \{\{\varnothing\}\}$.

❖ **Solution**
False

Set $\{\{\varnothing\}\}$ has exactly one element, which is $\{\varnothing\}$. But $\{\varnothing\} \neq \varnothing$.

Always–Sometimes–Never questions

In Questions 2.E.27 to 2.E.42, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

preview

2.E.27. Let a and b be arbitrary objects. Then $\{a,b\} = \{b,a\}$.

❖ **Solution**
Always

Sets ignore repetition and order, so $\{a,b\} = \{b,a\}$.

2.E.28. Let a, b and c be arbitrary objects. Then $\{a,b\} = \{a,c\}$.

❖ **Solution**
Sometimes

If $b = c$, then $\{a,b\} = \{a,c\}$.

If $b \neq c$, then $\{a,b\} \neq \{a,c\}$.

2.E.29. Let X and Y be sets and suppose there is some a such that $a \in X \Leftrightarrow a \in Y$. Then $X = Y$.

❖ **Solution**
Sometimes

There is some a such that $a \in X \Leftrightarrow a \in Y$ is not enough to deduce that $X = Y$. Set equality requires for every a

$$X = Y \iff \forall a, a \in X \Leftrightarrow a \in Y$$

For example, let $X = \{1\}, Y = \{1,2\}$, and let $a = 1$. Then $1 \in X \Leftrightarrow 1 \in Y$, but $X \neq Y$.

2.E.30. Let X and Y be sets and suppose there is some a such that $a \in X$ but $a \notin Y$. Then $X = Y$.

❖ **Solution**
Never

$a \in X$ but $a \notin Y$ means $X \nsubseteq Y$, double containment argument cannot hold. Therefore, X cannot be equal to Y.

2.E.31. Let X and Y be sets with $X \neq Y$. Then $\forall a,\, a \in X \Rightarrow a \notin Y$.

❖ **Solution**
Sometimes

If $X \cap Y = \varnothing$, this proposition is true.

If $X \cap Y$ is inhabited, this means there exists at least one element e such that $e \in X$ and $e \in Y$.

2.E.32. Let X and Y be sets with $X \cap Y = \varnothing$. Then $\forall a,\, a \in X \Rightarrow a \notin Y$.

❖ **Solution**
Always

If $X \cap Y = \varnothing$, this means for all elements in X, they are not in Y. i.e. $\forall a \in X, x \notin Y$. Therefore,

$$X \cap Y = \varnothing \implies \forall a,\, a \in X \Rightarrow a \notin Y$$

preview

2.E.33. Let X and Y be sets with $X \neq Y$. Then $\exists a, a \in X \wedge a \notin Y$.

❖ **Solution**
Sometimes

From $X \neq Y$ we have three cases:

$$X \nsubseteq Y \quad \text{or} \quad Y \nsubseteq X \quad \text{or} \quad \text{both}$$

If It happens to be the case

$$(X \subseteq Y) \wedge (Y \nsubseteq X) \implies X \neq Y$$

Then we have $X \subseteq Y \implies \forall a \in X, a \in Y$, i.e. $\nexists a, a \in X \wedge a \notin Y$.

In the other two cases, the conclusion is true

2.E.34. Let X and Y be sets of sets and suppose that $X \in Y$. Then $X \subseteq Y$.

❖ **Solution**
Sometimes

If $X = \varnothing$ and $Y = \{\varnothing\}$, then $X \in Y$. Since $\varnothing$ is a subset for all sets, then $X \subseteq Y$. The proposition is true.

If $X = \{\varnothing\}$ and $Y = \{\{\varnothing\}\}$, then $X \in Y$. But $X \nsubseteq Y$. The proposition is false.

2.E.35. Let X be a set. Then $\varnothing \in \mathscr{P}(X)$.

❖ **Solution**
Always

Since $\varnothing$ is a subset for all sets, it is necessarily an element of power set $\mathscr{P}(X)$. So $\varnothing \in \mathscr{P}(X)$ always be true.

2.E.36. Let X be a set. Then $\{\varnothing\} \in \mathscr{P}(X)$.

❖ **Solution**
Sometimes

If $\varnothing \in X$, then $\{\varnothing\} \in \mathscr{P}(X)$.

If $\varnothing \notin X$, then $\{\varnothing\} \notin \mathscr{P}(X)$.

2.E.37. Let X be a set. Then $\{\varnothing, X\} \subseteq \mathscr{P}(X)$.

❖ **Solution**
Always

Since $\varnothing$ is a subset of all sets, so $\varnothing \subset X$. Since any set is a subset of itself, so $X \subseteq X$.

Thus $\varnothing$ and X are two elements of $\mathscr{P}(X)$. Therefore $\{\varnothing, X\} \subseteq \mathscr{P}(X)$.

2.E.38. Let X and Y be sets such that $\mathscr{P}(X) = \mathscr{P}(Y)$. Then $X = Y$.

❖ **Solution**

Always

Suppose $\mathscr{P}(X) = \mathscr{P}(Y)$. Since any set is a subset of itself, so $X \in \mathscr{P}(X)$. Thus $X \in \mathscr{P}(Y)$. By definition of power set, we have

$$X \subseteq Y$$

Similarly, $Y \in \mathscr{P}(Y) = \mathscr{P}(X)$, we have

$$Y \subseteq X$$

By double containment, we have $X = Y$.

2.E.39. Let X and Y be sets such that $X \setminus Y = \varnothing$. Then $X = Y$.

❖ **Solution**

Sometimes

$X \setminus Y = \varnothing$ means there is no element that is in X but not in Y, i.e. $X \subseteq Y$. However, it does not imply $Y \subseteq X$. For example, let $X = \{1\}$ and $Y = \{1,2\}$, then $X \setminus Y = \varnothing$, but $X \neq Y$.

This proposition is true only if $Y \subseteq X$ holds additionally. then.

2.E.40. Let X, Y and Z be sets such that $X \setminus Y = Z$ and $Y \subseteq Z$. Then $X = Y \cup Z$.

❖ **Solution**

Always

Since $X \setminus Y = Z$, this means every element of Z is not in Y, i.e. $Z \cap Y = \varnothing$.

But we are given $Y \subseteq Z$, this forces Y must be $\varnothing$.Then

$$Z = X \setminus \varnothing = X$$

Thus

$$Y \cup Z = \varnothing \cup X = X$$

Therefore, $X = Y \cup Z$.

2.E.41. Let X and Y be sets. Then $X \cup Y = X \cap Y$.

❖ **Solution**

Sometimes

- If $X = Y$, then $X \cup Y = X \cap Y$.
  - ◇ ($\subseteq$) If $a \in X \cup Y$, this means $a \in X$ or $a \in Y$. Since $X = Y$, we have $a \in X$ and $a \in Y$, thus $a \in X \cap Y$. So $X \cup Y \subseteq X \cap Y$.
  - ◇ ($\supseteq$) If $a \in X \cap Y$, then $a \in X$ and $a \in Y$, thus $a \in X \cup Y$. So $X \cap Y \subseteq X \cup Y$.

  By double containment, we have $X = Y \implies X \cup Y = X \cap Y$.

preview

- If $X \neq Y$, then $X \cup Y \neq X \cap Y$.

  Since $X \neq Y$, there exists at least one element $a \in X$ but $a \notin Y$ or vice versa. According to the definition of intersection and union, $a \in X \cup Y$ but $a \notin X \cap Y$. Therefore, $X \cup Y \neq X \cap Y$.

  For example, let $X = \{1\}$ and $Y = \{1,2\}$, then

  $$X \cup Y = \{1,2\} \quad \text{and} \quad X \cap Y = \{1\}$$

  Clearly $X \cup Y \neq X \cap Y$.

2.E.42. Let X be a set and let $A \subseteq X$. Then $A \in X$.

❖ **Solution**

Sometimes

This question is similar to Question 2.E.34.

If $X = \varnothing$ and $Y = \{\varnothing\}$, the proposition is true. Since $\varnothing$ is a subset for all sets, then $X \subseteq Y$. $\varnothing$ is clearly an element of $\{\varnothing\}$, so $X \in Y$.

If $X = \{1\}$ and $Y = \{1,2\}$, then $X \subseteq Y$. But $X \notin Y$. The proposition is false.

preview

preview

Chapter 3

Functions

preview

Section 3.E

Chapter 3 exercises

3.E.1. For each of the following equations, determine whether there exists a function $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$, the equation holds if and only if $y = f(x)$.

(a) $x + y = 1$

(b) $x^2 + y^2 = 1$

(c) $x = 0$

(d) $y = 0$

(e) $(1 + x^2)y = 1$

(f) $(1 - x^2)y = 0$

3.E.2. Let X be a set. Prove that

$$\forall a \in X, \exists! U \in \mathscr{P}(X), (a \in U \wedge \exists! x \in X, x \in U)$$

Give an explicit description of the function $X \to \mathscr{P}(X)$ that is suggested by this logical formula.

3.E.3. Show that there is only one function whose codomain is empty. What is its domain?

Definition 3.E.4

A function $f : \mathbb{R} \to \mathbb{R}$ is **even** if $f(-x) = f(x)$ for all $x \in \mathbb{R}$, and it is **odd** if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$.

3.E.5. Let $n \in \mathbb{N}$. Prove that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^n$ for all $x \in \mathbb{R}$ is even if and only if n is even, and odd if and only if n is odd.

3.E.6. Prove that there is a unique function $f : \mathbb{R} \to \mathbb{R}$ that is both even and odd.

3.E.7. Let $U \subseteq \mathbb{R}$, and let $\chi_U : \mathbb{R} \to [2]$ be the indicator function of U.

(a) Prove that χ_U is an even function if and only if $U = \{-u \mid u \in U\}$;

(b) Prove that χ_U is an odd function if and only if $\mathbb{R} \setminus U = \{-u \mid u \in U\}$.

3.E.8. Prove that for every function $f : \mathbb{R} \to \mathbb{R}$, there is a unique even function $g : \mathbb{R} \to \mathbb{R}$ and a unique odd function $h : \mathbb{R} \to \mathbb{R}$ such that $f(x) = g(x) + h(x)$ for all $x \in \mathbb{R}$.

3.E.9. Let $\{\theta_n : [n] \to [n] \mid n \in \mathbb{N}\}$ be a family of functions such that $f \circ \theta_m = \theta_n \circ f$ for all $f : [m] \to [n]$. Prove that $\theta_n = \mathrm{id}_{[n]}$ for all $n \in \mathbb{N}$.

3.E.10. Let X be a set and let $U, V \subseteq X$. Describe the indicator function $\chi_{U \triangle V}$ of the symmetric difference of U and V (see Definition 2.E.8) in terms of χ_U and χ_V.

3.E.11. [This question assumes some familiarity with integral calculus.] A lie that is commonly told to calculus students is that

$$\int \frac{1}{x}\, dx = \log(|x|) + c$$

where log denotes the natural logarithm function and c is an arbitrary real constant. Prove that it is actually the case that

$$\int \frac{1}{x}\, dx = \log(|x|) + a\chi_U(x) + b\chi_V(x)$$

where a and b are arbitrary real constants, and U and V are particular (inhabited) subsets of $\mathbb{R} \setminus \{0\}$ that you should determine.

preview

Images and preimages

In Questions 3.E.12 to 3.E.15, find the image $f[U]$ of the subset U of the domain of the function f described in the question.

3.E.12. $f : \mathbb{R} \to \mathbb{R}$; $f(x) = \sqrt{1+x^2}$ for all $x \in \mathbb{R}$; $U = \mathbb{R}$.

3.E.13. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$; $f(a,b) = a + 2b$ for all $(a,b) \in \mathbb{Z} \times \mathbb{Z}$; $U = \{1\} \times \mathbb{Z}$.

3.E.14. $f : \mathbb{N} \to \mathscr{P}(\mathbb{N})$; $f(0) = \varnothing$ and $f(n+1) = f(n) \cup \{n\}$ for all $n \in \mathbb{N}$; $U = \mathbb{N}$.

3.E.15. $f : \mathbb{R}^{\mathbb{R}} \to \mathbb{R}^{\mathbb{R}}$ (where $\mathbb{R}^{\mathbb{R}}$ is the set of all functions $\mathbb{R} \to \mathbb{R}$); $f(h)(x) = h(|x|)$ for all $h \in \mathbb{R}^{\mathbb{R}}$ and all $x \in \mathbb{R}$; $U = \mathbb{R}^{\mathbb{R}}$.

In Questions 3.E.16 to 3.E.18, find the preimage $f^{-1}[V]$ of the subset V of the codomain of the function f described in the question.

3.E.16. $f : \mathbb{R} \to \mathbb{R}$; $f(x) = \sqrt{1+x^2}$ for all $x \in \mathbb{R}$; $V = (-5, 5]$.

3.E.17. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$; $f(a,b) = a + 2b$ for all $(a,b) \in \mathbb{Z} \times \mathbb{Z}$; $V = \{n \in \mathbb{Z} \mid n \text{ is odd}\}$.

3.E.18. $f : \mathscr{P}(\mathbb{N}) \times \mathscr{P}(\mathbb{N}) \to \mathscr{P}(\mathbb{N})$; $f(A,B) = A \cap B$ for all $(A,B) \in \mathscr{P}(\mathbb{N}) \times \mathscr{P}(\mathbb{N})$; $V = \{\varnothing\}$.

3.E.19. Let $f : X \to Y$ be a function. For each of the following statements, either prove it is true or find a counterexample.

(a) $U \subseteq f^{-1}[f[U]]$ for all $U \subseteq X$;

(b) $f^{-1}[f[U]] \subseteq U$ for all $U \subseteq X$;

(c) $V \subseteq f[f^{-1}[V]]$ for all $V \subseteq Y$;

(d) $f[f^{-1}[V]] \subseteq V$ for all $V \subseteq Y$.

3.E.20. Let $f : X \to Y$ be a function, let A be a set, and let $p : X \to A$ and $i : A \to Y$ be functions such that the following conditions hold:

(i) i is injective;

(ii) $i \circ p = f$; and

(iii) If $q : X \to B$ and $j : B \to Y$ are functions such that j is injective and $j \circ q = f$, then there is a unique function $u : A \to B$ such that $j \circ u = i$.

Prove that there is a unique bijection $v : A \to f[X]$ such that $i(a) = v(a)$ for all $a \in f[X]$.

3.E.21. Let $f : X \to Y$ be a function and let $U, V \subseteq Y$. Prove that:

(a) $f^{-1}[U \cap V] = f^{-1}[U] \cap f^{-1}[V]$;

(b) $f^{-1}[U \cup V] = f^{-1}[U] \cup f^{-1}[V]$; and

(c) $f^{-1}[Y \setminus U] = X \setminus f^{-1}[U]$.

Thus preimages preserve the basic set operations.

3.E.22. Let $f : X \to Y$ and $g : Y \to Z$ be functions.

(a) Prove that $(g \circ f)[U] = g[f[U]]$ for all $U \subseteq X$;

(b) Prove that $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$ for all $W \subseteq Z$.

preview

3.E.23. Let $f : X \to Y$ and $g : Y \to Z$ be functions. Define

$$S = \{(x,y,z) \in X \times Y \times Z \mid y = f(x) \text{ and } z = g(y)\}$$

and let $p : S \to X \times Z$ be the function defined by $p(x,y,z) = (x,z)$ for all $(x,y,z) \in S$. Prove that $p[S] = \mathrm{Gr}(g \circ f)$.

Injections, surjections and bijections

3.E.24. (a) Prove that, for all functions $f : X \to Y$ and $g : Y \to Z$, if $g \circ f$ is bijective, then f is injective and g is surjective.

(b) Find an example of a function $f : X \to Y$ and a function $g : Y \to Z$ such that $g \circ f$ is bijective, f is not surjective and g is not injective.

3.E.25. For each of the following pairs (U,V) of subsets of $\mathbb{R}$, determine whether the specification '$f(x) = x^2 - 4x + 7$ for all $x \in U$' defines a function $f : U \to V$ and, if it does, determine whether f is injective and whether f is surjective.

(a) $U = \mathbb{R}$ and $V = \mathbb{R}$;

(b) $U = (1,4)$ and $V = [3,7)$;

(c) $U = [3,4)$ and $V = [4,7)$;

(d) $U = (3,4]$ and $V = [4,7)$;

(e) $U = [2,\infty)$ and $V = [3,\infty)$;

(f) $U = [2,\infty)$ and $V = \mathbb{R}$.

3.E.26. For each of the following pairs of sets X and Y, find (with proof) a bijection $f : X \to Y$.

(a) $X = \mathbb{Z}$ and $Y = \mathbb{N}$;

(b) $X = \mathbb{R}$ and $Y = (-1,1)$;

(c) $X = [0,1]$ and $Y = (0,1)$;

(d) $X = [a,b]$ and $Y = (c,d)$, where $a,b,c,d \in \mathbb{R}$ with $a < b$ and $c < d$.

3.E.27. Prove that the function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined by $f(a,b) = \binom{a+b+1}{2} + b$ for all $(a,b) \in \mathbb{N} \times \mathbb{N}$ is a bijection.

3.E.28. Let $e : X \to X$ be a function such that $e \circ e = e$. Prove that there exist a set Y and functions $f : X \to Y$ and $g : Y \to X$ such that $g \circ f = e$ and $f \circ g = \mathrm{id}_Y$.

True–False questions

In Questions 3.E.29 to 3.E.36, determine (with proof) whether the statement is true or false.

3.E.29. The set $G = \{(x,y) \in \mathbb{R} \times \mathbb{R} \mid x^2 = y^2\}$ is the graph of a function.

3.E.30. The set $G = \{(x,y) \in \mathbb{Z} \times \mathbb{N} \mid x^2 = y^2\}$ is the graph of a function.

3.E.31. Every function with empty domain has an empty codomain.

preview

3.E.32. Every function with empty codomain has an empty domain.

3.E.33. The image of a function is a subset of its domain.

3.E.34. Given a function $f : X \to Y$, the assignment $V \mapsto f^{-1}[V]$ defines a function $\mathscr{P}(Y) \to \mathscr{P}(X)$.

3.E.35. Let f, g and h be composable functions. If $h \circ g \circ f$ is injective, then g is injective.

3.E.36. Every left inverse is surjective and every right inverse is injective.

Always–Sometimes–Never questions

In Questions 3.E.37 to 3.E.48, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

3.E.37. Let $f : X \to Y$ be a function and let $U \subseteq X$. Then there is a function $g : U \to Y$ defined by $g(x) = f(x)$ for all $x \in U$.

3.E.38. Let $f : X \to Y$ be a function and let $V \subseteq Y$. Then there is a function $g : X \to V$ defined by $g(x) = f(x)$ for all $x \in X$.

3.E.39. Let X be a set. Then there is a unique function $X \to \{0\}$.

3.E.40. Let X be a set. Then there is a unique function $X \to \varnothing$.

3.E.41. Let X and Y be sets and let $G \subseteq X \times Y$. Then G is the graph of a function $f : X \to Y$.

3.E.42. Let $f : \{1,2,3\} \to \{1,2\}$, let $G = \mathrm{Gr}(f)$ and let $G' = \{(y,x) \mid (x,y) \in G\}$. Then G' is the graph of a function $\{1,2\} \to \{1,2,3\}$.

3.E.43. Let $f : X \to Y$ be a function and let $U \subseteq X$ be inhabited. Then $f[U]$ is inhabited.

3.E.44. Let $f : X \to Y$ be a function and let $V \subseteq Y$ be inhabited. Then $f^{-1}[V]$ is inhabited.

3.E.45. Let $f : \{1,2\} \to \{1,2,3\}$ be a function. Then f is injective.

3.E.46. Let $f : \{1,2\} \to \{1,2,3\}$ be a function. Then f is surjective.

3.E.47. Let $a,b,c,d \in \mathbb{R}$ and define $f : \mathbb{R}^2 \to \mathbb{R}^2$ by $f(x,y) = (ax+by, cx+dy)$ for all $(x,y) \in \mathbb{R}^2$. Then f is injective if and only if f is surjective.

3.E.48. Let $U, V \subseteq \mathbb{R}$ and suppose that $x^2 \in V$ for all $x \in U$. The function $f : U \to V$ defined by $f(x) = x^2$ is injective.

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Chapter 4

Mathematical induction

preview

Section 4.E

Chapter 4 exercises

Recursive definitions

In Questions 4.E.1 to 4.E.5, use the recursive definitions of addition, multiplication and exponentiation directly to prove the desired equation.

4.E.1. $1+3=4$

4.E.2. $0+5=5$

4.E.3. $2\cdot 3=6$

4.E.4. $0\cdot 5=0$

4.E.5. $2^3=8$

4.E.6. Give a recursive definition of new quantifiers $\exists^{=n}$ for $n\in\mathbb{N}$, where given a set X and a predicate $p(x)$, the logical formula $\exists^{=n}x\in X,\ p(x)$ means 'there are exactly n elements $x\in X$ such that $p(x)$ is true'. That is, define $\exists^{=0}$, and then define $\exists^{=n+1}$ in terms of $\exists^{=n}$.

4.E.7. Use the recursive definition of binomial coefficients (**??**) to prove directly that $\binom{4}{2}=6$.

4.E.8. (a) Find the number of trailing 0s in the decimal expansion of 41!.

(b) Find the number of trailing 0s in the binary expansion of 41!.

4.E.9. Let N be a set, let $z\in N$ and let $s:N\to N$. Prove that (N,z,s) is a notion of natural numbers (in the sense of **??**) if and only if, for every set X, every element $a\in X$ and every function $f:X\to X$, there is a unique function $h:N\to X$ such that $h(z)=a$ and $h\circ f=s\circ h$.

Proofs by induction

4.E.10. Find all natural numbers n such that $n^5<5^n$.

4.E.11. Prove that $(1+x)^{123\,456\,789}\geqslant 1+123\,456\,789\,x$ for all real $x\geqslant -1$.

4.E.12. Let $a\in\mathbb{N}$ and assume that the last digit in the decimal expansion of a is 6. Prove that the last digit in the decimal expansion of a^n is 6 for all $n\geqslant 1$.

4.E.13. Let $a,b\in\mathbb{R}$, and let $a_0,a_1,a_2,\ldots$ be a sequence such that $a_0=a$, $a_1=b$ and $a_n=\dfrac{a_{n-1}+a_{n+1}}{2}$ for all $n\geqslant 1$. Prove that $a_n=a+(b-a)n$ for all $n\in\mathbb{N}$.

4.E.14. Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{R}$.

(a) Prove by induction that there is a real number a such that $f(n)=an$ for all $n\in\mathbb{N}$.

(b) Deduce that $f(n)=an$ for all $n\in\mathbb{Z}$.

(c) Deduce further that $f(x)=ax$ for all $x\in\mathbb{Q}$.

preview

4.E.15. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(0) > 0$ and $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Prove that there is some positive real number a such that $f(x) = a^x$ for all *rational* numbers x..

4.E.16. Let $a, b \in \mathbb{Z}$. Prove that $b - a$ divides $b^n - a^n$ for all $n \in \mathbb{N}$.

4.E.17. Prove by induction that $7^n - 2 \cdot 4^n + 1$ is divisible by 18 for all $n \in \mathbb{N}$.

Some examples from calculus

Questions 4.E.18 to 4.E.23 assume familiarity with the basic techniques of differential and integral calculus.

4.E.18. Prove that $\dfrac{d^n}{dx^n}(xe^x) = (n+x)e^x$ for all $n \in \mathbb{N}$.

4.E.19. Find and prove an expression for $\dfrac{d^n}{dx^n}(x^2e^x)$ for all $n \in \mathbb{N}$.

4.E.20. Let f and g be differentiable functions. Prove that

$$\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x)$$

for all $n \in \mathbb{N}$, where the notation $h^{(r)}$ denotes the rth derivative of h.

4.E.21. Find and prove an expression for the nth derivative of $\log(x)$ for all $n \in \mathbb{N}$.

4.E.22. Prove that $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ for all $\theta \in \mathbb{R}$ and all $n \in \mathbb{N}$.

4.E.23. (a) Prove that $\displaystyle\int_0^{\frac{\pi}{2}} \sin^{n+2}(x)\,dx = \frac{n+1}{n+2}\int_0^{\frac{\pi}{2}} \sin^n(x)\,dx$ for all $n \in \mathbb{N}$.

(b) Use induction to prove that $\displaystyle\int_0^{\frac{\pi}{2}} \sin^{2n}(x)\,dx = \frac{\pi}{2^{2n+1}}\binom{2n}{n}$ for all $n \in \mathbb{N}$.

(c) Find and prove an expression for $\displaystyle\int_0^{\frac{\pi}{2}} \sin^{2n+1}(x)\,dx$ for all $n \in \mathbb{N}$.

True–False questions

In Questions 4.E.24 to 4.E.32, determine (with proof) whether the statement is true or false.

4.E.24. Every factorial is positive.

4.E.25. Every binomial coefficient is positive.

4.E.26. Every proof by strong induction can be turned into a proof by weak induction by changing only the induction hypothesis.

4.E.27. Every proof by weak induction can be turned into a proof by strong induction by changing only the induction hypothesis.

4.E.28. Given a set X, a surjection $f : \mathbb{N} \to X$ and a logical formula $p(x)$ with free variable $x \in X$, if $p(f(0))$ is true and $\forall n \in \mathbb{N}$, $(p(f(n)) \Rightarrow p(f(n+1)))$ is true, then $\forall x \in X$, $p(x)$ is true.

preview

4.E.29. Given some logical formula $p(x)$ with free variable $x \in \mathbb{N}$, if for all $x \in \mathbb{N}$ there exists some $y \leqslant x$ such that $p(x)$ is false, then $p(x)$ is false for all $x \in \mathbb{N}$.

4.E.30. Every inhabited subset of the set $\mathbb{Q}^{\geqslant 0}$ of nonnegative rational numbers has a least element.

4.E.31. Every inhabited subset of the set $\mathbb{Z}$ of integers has a least element.

4.E.32. Every inhabited subset of the set $\mathbb{Z}^{-}$ of negative integers has a greatest element.

Always–Sometimes–Never questions

In Questions 4.E.33 to 4.E.35, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

4.E.33. Let $n,k,\ell \in \mathbb{Z}$. Then $\binom{n}{k+\ell} = \binom{n}{k} + \binom{n}{\ell}$.

4.E.34. Let $m,n \geqslant 2$. Then $(m+n)! = m! + n!$.

4.E.35. Let $p(x)$ be a logical formula with free variable $x \in \mathbb{Z}$, and suppose that $p(0)$ is true and, for all $n \in \mathbb{Z}$, if $p(n)$ is true, then $p(n+1)$ and $p(n-2)$ are true. Then $\forall n \in \mathbb{Z}$, $p(n)$ is true.

preview

Chapter 5

Relations

preview

Section 5.E

Chapter 5 exercises

Properties of relations

5.E.1. For each of the eight subsets

$$P \subseteq \{\text{reflexive}, \text{symmetric}, \text{transitive}\}$$

find a relation satisfying (only) the properties in P.

5.E.2. Prove that if R is a symmetric, antisymmetric relation on a set X, then it is a subrelation of the equality relation—that is, $\mathrm{Gr}(R) \subseteq \mathrm{Gr}(=)$.

5.E.3. A relation R on a set X is **left-total** if for all $x \in X$, there exists some $y \in X$ such that $x\,R\,y$. Prove that every left-total, symmetric, transitive relation is reflexive.

Equivalence relations

Definition 5.E.4

Let R be a relation on a set X and let $f : X \to Y$ be a function. The **transport** of R along f is the relation S on Y defined for $c, d \in Y$ by letting $c\,S\,d$ if and only if there exist $a, b \in X$ such that $f(a) = c$, $f(b) = d$ and $a\,R\,b$. That is

$$\mathrm{Gr}(S) = \{(f(a), f(b)) \mid a, b \in X,\ a\,R\,b\}$$

5.E.5. Let X and Y be sets and let $f : X \to Y$. Prove that if $\sim$ is an equivalence relation on X, then the transport of $\sim$ along f is symmetric and transitive. Under what condition(s) on f is the transport of $\sim$ along f an equivalence relation on Y?

Definition 5.E.6

Let R be any relation on a set X. The **equivalence relation generated by** R is the relation $\sim_R$ on X defined as follows. Given $x, y \in X$, say $x \sim_R y$ if and only if for some $k \in \mathbb{N}$ there is a sequence $(a_0, a_1, \ldots, a_k)$ of elements of X such that $a_0 = x$, $a_k = y$ and, for all $i \in [k]$, either $a_i R a_{i+1}$ or $a_{i+1} R a_i$.

5.E.7. Fix $n \in \mathbb{Z}$ and let R be the relation on $\mathbb{Z}$ defined by $x R y$ if and only if $y = x + n$. Prove that $\sim_R$ is the relation of congruence modulo n.

5.E.8. Let X be a set and let R be the subset relation on $\mathscr{P}(X)$. Prove that $U \sim_R V$ for all $U, V \subseteq X$.

5.E.9. Let X be a set, fix two distinct elements $a, b \in X$, and define a relation R on X by declaring $a R b$ only—that is, for all $x, y \in X$, we have $x R y$ if and only if $x = a$ and $y = b$. Prove that the relation $\sim_R$ is defined by $x \sim_R y$ if and only if either $x = y$ or $\{x, y\} = \{a, b\}$.

In Questions 5.E.10 to 5.E.13, let R be a relation on a set X, and let $\sim_R$ be the equivalence relation generated by R (as in Definition 5.E.6). In these questions, you will prove that $\sim_R$ is the 'smallest' equivalence relation extending R.

preview

5.E.10. Prove that $\sim_R$ is an equivalence relation on X.

5.E.11. Prove that $xRy \Rightarrow x \sim_R y$ for all $x, y \in X$.

5.E.12. Prove that if $\approx$ is any equivalence relation on X and $xRy \Rightarrow x \approx y$ for all $x, y \in X$, then $x \sim_R y \Rightarrow x \approx y$ for all $x, y \in X$.

5.E.13. Prove that if R is an equivalence relation, then $\sim_R = R$.

5.E.14. This question requires some familiarity with concepts from calculus. Let $\mathscr{C}^1(\mathbb{R})$ denote the set of all differentiable functions $F : \mathbb{R} \to \mathbb{R}$ whose derivative $F' : \mathbb{R} \to \mathbb{R}$ is continuous, and define a relation $\sim$ on $\mathscr{C}^1(\mathbb{R})$ by letting $F \sim G$ if and only if $F' = G'$.

(a) Prove that $\sim$ is an equivalence relation on $\mathscr{C}^1(\mathbb{R})$.

(b) Let $F \in \mathscr{C}^1(\mathbb{R})$. Prove that

$$[F]_\sim = \{G \in \mathscr{C}^1(\mathbb{R}) \mid \exists c \in \mathbb{R}, \forall x \in \mathbb{R}, G(x) = F(x) + c\}$$

(c) Interpret the notion of an indefinite integral $\int f(x)\,dx$ of a continuous function $f : \mathbb{R} \to \mathbb{R}$ in terms of $\sim$-equivalence classes.

True–False questions

In Questions 5.E.15 to 5.E.21, determine (with proof) whether the statement is true or false.

5.E.15. Every reflexive relation is symmetric.

5.E.16. There is a unique relation on $\mathbb{N}$ that is reflexive, symmetric and antisymmetric.

5.E.17. Every relation that is not symmetric is antisymmetric.

5.E.18. Every symmetric, antisymmetric relation is reflexive.

5.E.19. The empty set is the graph of a relation on $\mathbb{N}$.

5.E.20. For all sets X, the graph of a function $X \to X$ is the graph of a relation on X.

5.E.21. For all sets X, the graph of a relation on X is the graph of a function $X \to X$.

Always–Sometimes–Never questions

In Questions 5.E.22 to 5.E.23, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

5.E.22. Let $\sim$ be an equivalence relation on a set X. Then the relation $\approx$ on $\mathscr{P}(X) \setminus \{\varnothing\}$, defined by $A \approx B$ if and only if $x \sim y$ for all $x \in A$ and $y \in B$, is an equivalence relation.

5.E.23. Let $\sim$ be an equivalence relation on a set X. Then the relation $\approx$ on $\mathscr{P}(X) \setminus \{\varnothing\}$, defined by $A \approx B$ if and only if $x \sim y$ for some $x \in A$ and $y \in B$, is an equivalence relation.

preview

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Chapter 6

Finite and infinite sets

preview

Section 6.E

Chapter 6 exercises

Finite sets

6.E.1. Let $n \in \mathbb{N}$ and let $f : [n] \to [n]$ be a function. Prove that f is injective if and only if f is surjective.

6.E.2. Prove that $|\mathbb{Z}/n\mathbb{Z}| = n$ for all integers $n > 0$.

In Questions 6.E.3 to 6.E.4 the symbol $\triangle$ denotes the*symmetric difference* operation defined in Definition 2.E.8.

6.E.3. Let A and B be sets.

(a) Prove that if A and B are finite, then $A \triangle B$ is finite, and express $|A \triangle B|$ in terms of $|A|$, $|B|$ and $|A \cap B|$.

(b) Prove that if $A \triangle B$ is finite then $A \setminus B$ and $B \setminus A$ are finite.

(c) Prove that there are infinite sets A and B such that $A \triangle B$ is finite.

6.E.4. Let A, B and C be sets. Prove that if $A \triangle B$ and $B \triangle C$ are finite, then $A \triangle C$ is finite, where the notation $\triangle$ refers to the *symmetric difference* operation defined in Definition 2.E.8.

True–False questions

In Questions 6.E.5 to 6.E.8, determine (with proof) whether the statement is true or false.

6.E.5. Every finite set has a unique enumeration.

6.E.6. The union of a finite number of finite sets is finite.

6.E.7. Every finite set is a proper subset of another finite set.

6.E.8. There is a finite set with infinitely many subsets.

Always–Sometimes–Never questions

In Questions 6.E.9 to 6.E.15, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

6.E.9. Let X and Y be finite sets and let $f : X \to Y$ be a function. If $|X| \leqslant |Y|$, then f is injective.

6.E.10. Let X be a finite set. Then there is some proper subset $U \subsetneqq X$ such that $|U| = |X|$.

6.E.11. Let X be an infinite set. Then $\mathbb{N} \subseteq X$.

6.E.12. Let X be a finite set. Then $\mathscr{P}(X)$ is finite.

preview

6.E.13. Let $a, b \in \mathbb{R}$ with $a < b$. Then the set $[a,b] \cap \mathbb{Q}$ is finite.

6.E.14. Let X and Y be sets with X finite, and let $f : X \to Y$ be an injection. Then Y is finite.

6.E.15. Let X and Y be sets with X finite, and let $f : X \to Y$ be a surjection. Then Y is finite.

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Part II

Topics in pure mathematics

preview

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Chapter 7

Number theory

preview

Section 7.E

Chapter 7 exercises

Greatest common divisors

In Questions 7.E.1 to 7.E.3, use the Euclidean algorithm (**??**) to find the greatest common divisor of the given pair of integers.

7.E.1. 382 and 218

7.E.2. 368475 and 26010

7.E.3. 24004512 and 10668672

7.E.4. Let $a,b,c,d \in \mathbb{Z}$ and suppose that $ad-bc=1$. Prove that $a+b$ and $c+d$ are coprime.

Definition 7.E.5

Let $f(x)$ and $g(x)$ be polynomials over $\mathbb{Z}$. We say $f(x)$ **divides** $g(x)$ if there exists a polynomial $q(x)$ over $\mathbb{Z}$ such that $g(x)=q(x)f(x)$.

7.E.6. Prove that $1-x$ divides $1-x^2$ and that $1-x$ does not divide $1+x^2$.

7.E.7. Prove that $1+x+x^2$ divides $1+x^4+x^5$.

Definition 7.E.8

Let $f(x)$ be a polynomial over $\mathbb{Z}$. The **degree** of f, written $\deg(f(x))$, is the greatest power of x in f whose coefficient is nonzero, unless f is the zero polynomial, whose degree is defined to be $-\infty$. We adopt the convention that $(-\infty)+r=-\infty=r+(-\infty)$ for all $r \in \mathbb{N}\cup\{-\infty\}$.

7.E.9. Let $f(x)$ and $g(x)$ be polynomials over $\mathbb{Z}$. Prove that $\deg(f(x),g(x)) = \deg(f(x)) + \deg(g(x))$.

7.E.10. Prove the following modified form of the division theorem (**??**) for polynomials: given polynomials $f(x)$ and $g(x)$ over $\mathbb{Z}$, with $f(x)\neq 0$, prove that there exist unique polynomials $q(x)$ and $r(x)$ over $\mathbb{Z}$ such that $g(x)=q(x)f(x)+r(x)$ and $\deg(r(x))<\deg(f(x))$.

Prime numbers

7.E.11. Prove that, for all $k \in \mathbb{N}$, the function $i:\mathbb{N}^k \to \mathbb{N}$ defined by

$$i(n_1,n_2,\ldots,n_k) = p_1^{n_1}p_2^{n_2}\cdots p_k^{n_k}$$

for all $(n_1,n_2,\ldots,n_k)\in\mathbb{N}^k$ is an injection, where $p_1,p_2,p_3\ldots$ is an enumeration of the set of positive primes in increasing order. (Thus $p_1=2$, $p_2=3$, $p_3=5$, and so on.)

7.E.12. Use the result of Question 7.E.11 to construct a bijection

$$\bigcup_{k\in\mathbb{N}} \left(\mathbb{N}^k \setminus \{(0,0,\ldots,0)\}\right) \to \{n\in\mathbb{N} \mid n \geqslant 2\}$$

preview

7.E.13. Use a method akin to that of Question 7.E.11 to define an injection $\mathbb{Z}^k \to \mathbb{N}$.

7.E.14. Define a subset $A \subseteq \mathbb{Z}$ by

$$A = \{n \in \mathbb{N} \mid \exists k > 0,\ n \mid 12^k - 1\}$$

Find all prime numbers in $\mathbb{Z} \setminus A$.

Base-b expansions

7.E.15. Let $n \in \mathbb{N}$. Prove that the number of trailing 0s in the decimal expansion of $n!$ is equal to

$$\sum_{k=1}^{d} \left\lfloor \frac{n}{5^k} \right\rfloor$$

where $d \in \mathbb{N}$ is least such that $5^{d+1} > n$, and where $\lfloor x \rfloor$ (LaTeX code: `\lfloor,\rfloor`) denotes the floor of x (see **??**).

7.E.16. Let $b \in \mathbb{N}$ with $b \geqslant 2$. Find an expression in terms of $n \in \mathbb{N}$ for the number of trailing 0s in the base-b expansion of $n!$.

True–False questions

In Questions 7.E.17 to 7.E.22, determine (with proof) whether the statement is true or false.

7.E.17. There is an integer that is not coprime to any integer.

7.E.18. Every linear Diophantine equation has a solution.

7.E.19. Every integer n is coprime to its successor $n+1$.

7.E.20. If the greatest common divisors of two integers is 3, and their least common multiple of 7, then their product is 21.

7.E.21. Every integer is congruent modulo 7 to its remainder when divided by 21.

7.E.22. Let $k \geqslant 1$. Then $15^k \equiv 1 \pmod 6$.

Always–Sometimes–Never questions

In Questions 7.E.23 to 7.E.31, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

7.E.23. Let $a, b, c \in \mathbb{Z}$ and suppose that $\gcd(a,b) = 2$ and $\gcd(b,c) = 2$. Then $\gcd(a,c) = 2$.

7.E.24. Let $a, b \in \mathbb{Z}$. Then $\gcd(a,b) = \gcd(a, \gcd(a,b))$.

7.E.25. Let $a, b \geqslant 2$. Then $\gcd(a,b) = \operatorname{lcm}(a,b)$.

7.E.26. Let $n \in \mathbb{Z}$. Then $n^2 + 1$ is coprime to $n^4 + n^2 + 1$.

preview

7.E.27. Let $p \in \mathbb{Z}$ be prime. Then p is coprime to all integers $a \neq p$.

7.E.28. Let $p, q, r, s \in \mathbb{Z}$ be positive primes and suppose that $pq = rs$. Then $p = r$ and $q = s$.

7.E.29. Let $p_1, p_2, \ldots, p_s, q_1, q_2, \ldots, q_t \in \mathbb{Z}$ be prime and suppose that $p_1 p_2 \ldots p_s = q_1 q_2 \ldots q_t$. Then $s = t$.

7.E.30. Let $a, b \in \mathbb{Z}$ and let p be a positive prime. Then $p \mid (a+b)^p - a - b$.

7.E.31. Let $n \geqslant 0$ and let $b > 0$. Then n is congruent modulo $b-1$ to the sum of the digits in its base-b expansion.

preview

Chapter 8

Enumerative combinatorics

preview

Section 8.E

Chapter 8 exercises

Counting

8.E.1. Let X and Y be finite sets with $|X| = m \in \mathbb{N}$ and $|Y| = n \in \mathbb{N}$. Prove that there are 2^{mn} relations from X to Y.

8.E.2. Let X be a set and let R be a relation on X. Prove that R is reflexive if and only if $\Delta_X \subseteq \mathrm{Gr}(R)$, where Δ_X is the diagonal subset of $X \times X$ (see **??**). Deduce that if X is finite and $|X| = n \in \mathbb{N}$, then there are $2^{n(n-1)}$ reflexive relations on X.

8.E.3. Let X be a finite set with $|X| = n \in \mathbb{N}$. Prove that there are $2^{\binom{n}{2}} \cdot 2^n$ symmetric relations on X.

8.E.4. Let X be a finite set with $|X| = n \in \mathbb{N}$. Prove that there are $3^{\binom{n}{2}} \cdot 2^n$ antisymmetric relations on X.

8.E.5. Let X be a finite set with $|X| = n \in \mathbb{N}$, let $\sim$ be an equivalence relation on X, and suppose that there is some natural number k such that $|[a]_\sim| = k$ for all $a \in X$. Prove that k divides n, and that $|X/\!\sim| = \dfrac{n}{k}$.

8.E.6. Let $n, k \in \mathbb{N}$ with $n > 0$ and $k \leqslant n$. Prove that the number of functions $f : [n] \to [n]$ that fix exactly k elements of $[n]$ is equal to $\binom{n}{k}(n-1)^{n-k}$.

Double counting

8.E.7. Let $a, b, m, n \in \mathbb{N}$. Prove each of the following by double counting.

(a) $a(m+n) = am + an$

(b) $a^{m+n} = a^m \cdot a^n$

(c) $(a^m)^n = a^{mn}$

(d) $(ab)^n = a^n \cdot b^n$

8.E.8. Prove that $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$ for all $n \in \mathbb{N}$

8.E.9. Prove that $\displaystyle\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m} = 2^{n-m}\binom{n}{m}$ for all $m, n \in \mathbb{N}$ with $m \leqslant n$.

8.E.10. Prove that $\displaystyle\sum_{j=0}^{k} \binom{n-j}{k-j} = \binom{n+1}{k}$ for all $n, k \in \mathbb{N}$ with $k \leqslant n$.

8.E.11. Prove that $\displaystyle\sum_{k=1}^{n}\sum_{\ell=0}^{k} k\binom{n}{k}\binom{n-k}{\ell} = n \cdot 3^{n-1}$ for all $n \in \mathbb{N}$.

8.E.12. Prove that $\displaystyle\binom{n}{r+s+1} = \sum_{k=r+1}^{n-s} \binom{k-1}{r}\binom{n-k}{s}$ for all $n, r, s \in \mathbb{N}$.

preview

8.E.13. Let $a_1, a_2, \ldots, a_r \in \mathbb{N}$ and let $n = a_1 + a_2 + \cdots + a_r$. Prove that

$$\binom{n}{a_1, a_2, \ldots, a_r} = \prod_{k=0}^{r-1} \binom{n - \sum_{i=1}^{k} a_i}{a_{k+1}}$$

where $\binom{n}{a_1, a_2, \ldots, a_r}$ is the number of ordered r-tuples $(U_1, U_2, \ldots, U_r)$ such that $U_1, U_2, \ldots, U_r$ is a partition of $[n]$ and $|U_k| = a_k$ for all $1 \leqslant k \leqslant r$.

Alternating sums

8.E.14. Let X be a finite set. Prove that if $|X|$ is odd then there is no parity-swapping involution $X \to X$.

8.E.15. Find the number of subsets of $[100]$ that do not contain a multiple of 8.

Miscellaneous exercises

8.E.16. Let $n \in \mathbb{N}$. For each $k \in \mathbb{N}$, find the coefficient of x^k in the polynomial $p(x)$ defined by $p(x) = (1 + x + x^2)^n$.

8.E.17. Fix $n \in \mathbb{N}$. For each $k \in \mathbb{N}$, find the coefficient of x^k in the polynomial $p(x)$ defined by $p(x) = \sum_{i=0}^{n} x^{n-i}(1+x)^i$.

True–False questions

In Questions 8.E.18 to 8.E.18, determine (with proof) whether the statement is true or false.

8.E.18. For every partition $\mathscr{A}$ of a finite set X, there is a function $f : X \to X$ that is a parity-swapping involution with respect to $\mathscr{A}$.

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Chapter 9

Real numbers

preview

Section 9.E

Chapter 9 exercises

Inequalities and means

9.E.1. Compute $\|2(-1,-5,1,5,1)-(2,-1,1,-2,8)\|$.

9.E.2. Let $a,b\in\mathbb{R}$ and let $a_0,a_1,a_2,\ldots,a_n\in\mathbb{R}$ with $a_0=a$ and $a_n=b$. Prove that $|a-b|\leqslant\sum_{k=1}^{n}|a_{k-1}-a_k|$.

9.E.3. Let $n\in\mathbb{N}$. Prove that $\vec{x}\cdot\vec{y}=\vec{y}\cdot\vec{x}$ for all $\vec{x},\vec{y}\in\mathbb{R}^n$.

9.E.4. Let $n\in\mathbb{N}$, let $\vec{x},\vec{y}\in\mathbb{R}^n$ and let $a,b,c,d\in\mathbb{R}$. Prove that $(a\vec{x}+b\vec{y})\cdot(c\vec{x}+d\vec{y})=ac\|\vec{x}\|^2+bd\|\vec{y}\|^2+(ad+bc)(\vec{x}\cdot\vec{y})$.

9.E.5. Let $a,b,c,d\in\mathbb{R}$. Assume that $a^2+b^2=9$, $c^2+d^2=25$ and $ac+bd=15$. Find the value of $\dfrac{a+2b}{c+2d}$.

9.E.6. Let $\vec{a}\in\mathbb{R}^3$ and $r>0$, and define

$$B(\vec{a};r)=\{\vec{x}\in\mathbb{R}^3\mid\|\vec{x}-\vec{a}\|<r\}$$

Prove that $\|\vec{x}-\vec{y}\|<2r$ for all $\vec{x},\vec{y}\in B(\vec{a};r)$.

9.E.7. Prove that $a^3+b^3+c^3\geqslant abc$ for all positive real numbers a,b,c. When does equality occur?

9.E.8. Let $a,b,c>0$. Prove that $\dfrac{b+c}{a}+\dfrac{c+a}{b}+\dfrac{a+b}{c}\geqslant 6$, with equality if and only if $a=b=c$.

9.E.9. Let $a,b,c\in\mathbb{R}$. Prove that $a^4+b^4+c^4\geqslant\dfrac{(ab+bc+ca)^2}{3}$. When does equality occur?

9.E.10. *Pound cost averaging* is an investment strategy in which an investor purchases asset in equal instalments over a fixed period of time.

Let $a,b\in\mathbb{R}$ with $a<b$, let $n\geqslant 1$, and let $f:[a,b]\to(0,\infty)$ be such that at time $t\in[a,b]$, a stock is trading at £$f(t)$ per share. Assume that you purchase a total of £M of the stock in equal instalments of £$\dfrac{M}{n}$ at times $t_0,t_1,\ldots,t_{n-1}$, where $M\in[0,\infty)$, $n\geqslant 1$, and $t_i=a+i\cdot\dfrac{b-a}{n}$ for all $0\leqslant i<n$.

(a) Prove that the value of the shares that you hold at time b is equal to

$$£\,\frac{f(b)}{\mathrm{HM}\big(f(t_0),\,f(t_1),\,\ldots,\,f(t_{n-1})\big)}\cdot M$$

where $\mathrm{HM}(a_1,a_2,\ldots,a_n)$ denotes the harmonic mean of $a_1,a_2,\ldots,a_n\in\mathbb{R}$.

(b) Under what condition have you made a profit?

preview

Sequences

9.E.11. Does there exist a sequence (x_n) such that $(x_{n+1} - x_n) \to 0$ but (x_n) diverges?

True–False questions

In Questions 9.E.12 to 9.E.18, determine (with proof) whether the statement is true or false.

9.E.12. For all $x, y \in \mathbb{R}$ we have $|x-y| \leqslant |x| - |y|$.

9.E.13. The function $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(\vec{x}) = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$ is injective.

9.E.14. The function $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(\vec{x}) = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$ is surjective.

9.E.15. For all $\vec{x}, \vec{y} \in \mathbb{R}^n$ we have $\|2\vec{x} - 3\vec{y}\| \leqslant 2\|\vec{x}\| + 3\|\vec{y}\|$.

9.E.16. Every convergent sequence is bounded.

9.E.17. Every convergent sequence is eventually monotone.

9.E.18. Every subsequence of a divergent sequence diverges.

Always–Sometimes–Never questions

In Questions 9.E.19 to 9.E.26, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

9.E.19. Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and suppose that $\|\vec{x} - \vec{y}\| = 0$. Then $\vec{x} = \vec{y}$.

9.E.20. Let $x_1, x_2, \ldots, x_n \geqslant 0$. Then the harmonic mean of the x_is is greater than their quadratic mean.

9.E.21. Let $y_1, y_2, \ldots, y_m \geqslant 0$. Then the geometric mean of the y_js is less than or equal to their quadratic mean.

9.E.22. Let (x_n) be a sequence of real numbers and suppose that the subsequences (x_{2n}) and (x_{2n+1}) both converge. Then (x_n) converges.

9.E.23. Let (x_n) be a sequence of real numbers and suppose that the subsequences (x_{2n}), (x_{2n+1}) and (x_{3n}) all converge. Then (x_n) converges.

9.E.24. Let $f : \mathbb{R} \to \mathbb{R}$, let (x_n) be a sequence of real numbers, and suppose that $(x_n) \to a \in \mathbb{R}$. Then $(x_n^2) \to a^2$.

9.E.25. Let $f : \mathbb{R} \to \mathbb{R}$, let (x_n) be a sequence of real numbers, and suppose that $(x_n) \to a \in \mathbb{R}$. Then $(f(x_n)) \to f(a)$.

9.E.26. Let (a_n) be a sequence of real numbers and suppose that $(a_n) \to 0$. Then $\sum_{n \geqslant 0} a_n$ converges.

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Chapter 10

Infinity

preview

Section 10.E

Chapter 10 exercises

Countability

In Questions 10.E.1 to 10.E.4, prove that the set is countable.

10.E.1. The set $D = \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\}$ of *dyadic* rational numbers.

10.E.2. The set of all functions $[n] \to \mathbb{Z}$, where $n \in \mathbb{N}$.

10.E.3. The set of all real numbers whose square is rational.

10.E.4. The following set:

$$[(\mathbb{Z} \times \mathbb{Q}) \setminus (\mathbb{N} \times \mathbb{Z})] \cup \{n \in \mathbb{N} \mid \exists u, v \in \mathbb{N}, n = 5u + 6v\} \cup \{x \in \mathbb{R} \mid x - \sqrt{2} \in \mathbb{Q}\}$$

In Questions 10.E.5 to 10.E.7, prove that the set is uncountable.

10.E.5. The set of all functions $\mathbb{Z} \to [2]$.

10.E.6. The set of all subsets $U \subseteq \mathbb{N}$ such that neither U nor $\mathbb{N} \setminus U$ is finite.

10.E.7. The set of all sequences of rational numbers that converge to 0.

In Questions 10.E.8 to 10.E.12, determine whether the set is countable or uncountable, and then prove it.

10.E.8. The set of all functions $f : \mathbb{N} \to \mathbb{N}$ that are weakly decreasing—that is, such that for all $m, n \in \mathbb{N}$, if $m \leqslant n$, then $f(m) \geqslant f(n)$.

10.E.9. The set of all functions $f : \mathbb{N} \to \mathbb{Z}$ that are weakly decreasing.

10.E.10. The set of all periodic functions $f : \mathbb{Z} \to \mathbb{Q}$—that is, such that there is some integer $p > 0$ such that $f(x + p) = f(x)$ for all $x \in \mathbb{Z}$.

10.E.11. The set of all periodic functions $f : \mathbb{Q} \to \mathbb{Z}$—that is, such that there is some rational number $p > 0$ such that $f(x + p) = f(x)$ for all $x \in \mathbb{Q}$.

10.E.12. The set of all real numbers x such that $a_0 + a_1 x + \cdots + a_d x^d = 0$ for some rational numbers $a_0, a_1, \ldots, a_d$ with $a_d \neq 0$.

10.E.13. A subset $D \subseteq \mathbb{R}$ is *dense* if $(a - \varepsilon, a + \varepsilon) \cap U$ is inhabited for all $a \in \mathbb{R}$ and all $\varepsilon > 0$—intuitively, this means that there are elements of U arbitrarily close to any real number. Must a dense subset of $\mathbb{R}$ be uncountable?

Cardinality

In Questions 10.E.14 to 10.E.20, determine whether the cardinality of the set is equal to $\aleph_0$, equal to $\mathfrak{c}$, or greater than $\mathfrak{c}$.

preview

10.E.14. $\mathscr{P}(\mathbb{R})$.

10.E.15. The set of all finite subsets of $\mathbb{R}$.

10.E.16. The set of all countably infinite subsets of $\mathbb{R}$.

10.E.17. $\mathbb{R} \times \mathbb{R}$.

10.E.18. The set of all functions $f : \mathbb{N} \to \mathbb{R}$ such that $f(n) = 0$ for all but finitely many $n \in \mathbb{N}$.

10.E.19. The set of all functions $\mathbb{Q} \to \mathbb{R}$.

10.E.20. The set of all functions $\mathbb{R} \to \mathbb{R}$.

10.E.21. Prove that there is a set $\mathscr{C}$ of pairwise disjoint circles in $\mathbb{R}^2$ such that $|\mathscr{C}| = \mathfrak{c}$; formally, a circle is a subset $C \subseteq \mathbb{R}^2$ of the form

$$C = \{(x,y) \mid (x-a)^2 + (y-b)^2 = r^2\}$$

for some $a, b \in \mathbb{R}$ and some $r > 0$.

10.E.22. Prove that there does not exist a set $\mathscr{D}$ of pairwise disjoint discs in $\mathbb{R}^2$ such that $|\mathscr{D}| = \mathfrak{c}$; formally, a disc is a subset $D \subseteq \mathbb{R}^2$ of the form

$$D = \{(x,y) \mid (x-a)^2 + (y-b)^2 \leqslant r^2\}$$

for some $a, b \in \mathbb{R}$ and some $r > 0$.

Cardinal arithmetic

10.E.23. Prove that $\kappa + 0 = \kappa \cdot 1 = \kappa^1 = \kappa$ for all cardinal numbers κ.

10.E.24. Prove that $\kappa^0 = 1^\kappa = 1$ for all cardinal numbers κ.

10.E.25. Let κ be an infinite cardinal number such that $\kappa \cdot \kappa = \kappa$. Prove that $\kappa^\kappa = 2^\kappa$.

10.E.26. Prove that $\mathfrak{c}^\mathfrak{c} = 2^{2^{\aleph_0}}$.

10.E.27. Let κ, λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leqslant \lambda$, then $\kappa + \mu \leqslant \lambda + \mu$.

(b) Suppose that $\kappa + \mu \leqslant \lambda + \mu$. Must it be the case that $\kappa \leqslant \lambda$?

10.E.28. Let κ, λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leqslant \lambda$, then $\kappa \cdot \mu \leqslant \lambda \cdot \mu$.

(b) Suppose that $\kappa \cdot \mu \leqslant \lambda \cdot \mu$ and $\mu > 0$. Must it be the case that $\kappa \leqslant \lambda$?

10.E.29. Let κ, λ and μ be cardinal numbers.

(a) Prove that if $\kappa \leqslant \lambda$, then $\kappa^\mu \leqslant \lambda^\mu$.

(b) Suppose that $\kappa^\mu \leqslant \lambda^\mu$ and $\mu > 0$. Must it be the case that $\kappa \leqslant \lambda$?

10.E.30. Let κ, λ and μ be cardinal numbers.

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(a) Prove that if $\kappa \leqslant \lambda$, then $\mu^\kappa \leqslant \mu^\lambda$.

(b) Suppose that $\mu^\kappa \leqslant \mu^\lambda$ and $\mu > 1$. Must it be the case that $\kappa \leqslant \lambda$?

Definition 10.E.31
Given cardinal numbers κ and λ, define the **binomial coefficient** $\binom{\kappa}{\lambda}$ by

$$\binom{\kappa}{\lambda} = |\{U \subseteq [\kappa] \mid |U| = \lambda\}|$$

10.E.32. Let κ be a cardinal number. Prove that

$$\binom{\aleph_0}{\kappa} = \begin{cases} 1 & \text{if } \kappa = 0 \\ \aleph_0 & \text{if } \kappa \in \mathbb{N} \text{ and } \kappa > 0 \\ 2^{\aleph_0} & \text{if } \kappa = \aleph_0 \\ 0 & \text{if } \kappa \notin \mathbb{N} \cup \{\aleph_0\} \end{cases}$$

10.E.33. Find the values of $\binom{\mathfrak{c}}{\kappa}$ for $\kappa \in \mathbb{N} \cup \{\aleph_0, \mathfrak{c}\}$.

10.E.34. Prove that $\sum_{\lambda \leqslant \kappa} \binom{\kappa}{\lambda} = 2^\kappa$ for all cardinal numbers κ.

10.E.35. Define the factorial $\kappa!$ of a cardinal number κ in a way that generalises the notion of factorial $n!$ for a natural number n. Find expressions for $\aleph_0!$ and $\mathfrak{c}!$.

True–False questions

In Questions 10.E.36 to 10.E.46, determine (with proof) whether the statement is true or false.

10.E.36. For any two countably infinite sets X and Y, there exists a bijection $X \to Y$.

10.E.37. For any two uncountable sets X and Y, there exists a bijection $X \to Y$.

10.E.38. The product of countably many countable sets is countable.

10.E.39. Every countably infinite set can be partitioned into infinitely many infinite subsets.

10.E.40. There are countably many infinite subsets of $\mathbb{N}$.

10.E.41. The set $\mathbb{N} \times \mathbb{Z} \times \mathbb{Q}$ is countably infinite.

10.E.42. $\aleph_0$ is the smallest cardinal number.

10.E.43. There exist sets X and Y of different cardinalities that have the same number of finite subsets.

10.E.44. There exist sets X and Y of different cardinalities that have the same number of countably infinite subsets.

10.E.45. For all sets X, if $U \subsetneqq X$, then $|U| < |X|$.

10.E.46. $2^\kappa < 3^\kappa$ for all cardinal numbers $\kappa > 0$.

preview

Always–Sometimes–Never questions

In Questions 10.E.47 to 10.E.52, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

10.E.47. Let X be a set and suppose that there exists an injection $\mathbb{N} \to X$. Then there exists an injection $\mathbb{Q} \to X$.

10.E.48. Let X be a set. Then $\mathscr{P}(X)$ is countably infinite.

10.E.49. Let X be a set, let κ and λ be cardinal numbers, and suppose that $\lambda \leqslant \kappa = |X|$. Then X has a subset of size λ.

10.E.50. Let X and Y be sets with $|X| \leqslant |Y|$. Then $|\mathscr{P}(X)| \leqslant |\mathscr{P}(Y)|$.

10.E.51. Let $n \in \mathbb{N}$ with $n \geqslant 2$. Then $\aleph_0^n > \aleph_0$.

10.E.52. Let κ be a cardinal number with $\kappa \geqslant \aleph_0$. Then $\aleph_0^\kappa > \aleph_0$.

Trick question

10.E.53. Does there exist a cardinal number κ such that $\aleph_0 < \kappa < 2^{\aleph_0}$?

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Chapter 11

Discrete probability theory

preview

Section 11.E

Chapter 11 exercises

True–False questions

In Questions 11.E.1 to 11.E.5, determine (with proof) whether the statement is true or false.

11.E.1. In a probability space $(\Omega, \mathbb{P})$, we have $\mathbb{P}(A \mid \Omega) = \mathbb{P}(A)$ for all events A.

11.E.2. The function $\mathbb{P} : \mathscr{P}(\mathbb{N}) \to [0,1]$ defined by $\mathbb{P}(A) = \sum_{n \in A} 2^{-n}$ is a probability measure on $\mathbb{N}$.

11.E.3. The function $\mathbb{P} : \mathscr{P}(\mathbb{Z}) \to [0,1]$ defined by $\mathbb{P}(A) = \sum_{n \in A} 2^{-n}$ is a probability measure on $\mathbb{Z}$.

11.E.4. The function $\mathbb{P} : \mathscr{P}(\mathbb{N}) \to [0,1]$, defined by letting $\mathbb{P}(\varnothing) = \varnothing$ and $\mathbb{P}(A) = 2^{-\min(A)}$ for all $A \neq \varnothing$, is a probability measure on $\mathbb{N}$.

11.E.5. A random variable can have infinite expectation.

Always–Sometimes–Never questions

In Questions 11.E.6 to 11.E.10, determine (with proof) whether the conclusion is always, sometimes or never true under the given hypotheses.

11.E.6. Let $(\Omega, \mathbb{P})$ be a probability space. The only event whose probability is zero is the empty event $\varnothing \subseteq \Omega$.

11.E.7. Let $(\Omega, \mathbb{P})$ be a probability space and let A be an event. Then $\mathbb{P}(A)^2 - \mathbb{P}(A) = 2$.

11.E.8. Let $(\Omega, \mathbb{P})$ be a probability space and let A and B be events with $\mathbb{P}(B) > 0$. Then $\mathbb{P}(A \mid B) \geqslant \mathbb{P}(A)$.

11.E.9. Let $(\Omega, \mathbb{P})$ be a probability space and let A and B be mutually exclusive events with $\mathbb{P}(B) > 0$. Then $\mathbb{P}(A \mid B) = 0$.

11.E.10. Let $(\Omega, \mathbb{P})$ be a probability space and let A, B and C be events with $\mathbb{P}(B) > 0$ and $\mathbb{P}(C) > 0$. Then $\mathbb{P}(A \mid B)\mathbb{P}(B \mid C) = \mathbb{P}(A \mid C)$.

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Chapter 12

Additional topics

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Section 12.E

Chapter 12 exercises

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